

Implementation Guide

Architecture notes, project conventions, implementation seams, and decision records.

Working draft

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Packet Runtime Modernization

Status

Pre-reseed modernization chapter status. The early sections preserve the chapter history; the current state has seeded Definition/Bundle packet material, closed in-scope runtime genericization, packet-based policy/dependency semantics, retired legacy bridge write intents, and trusted signed Preference.element writes.

Summary

This chapter brings the packet system, trusted runtime coordination, runtime connectors, and mutation orchestration up to the Preference-as-template direction. Definition, Bundle, and Preference are now canonical packet types; Preference remains the working runtime example with a signed trusted write path and runtime connector coverage for comparison. The broader goal is to keep every live packet type visible through the same coverage map while moving mutations into standardized Trusted Runtime Coordinator paths.

The work should preserve current behavior while replacing hidden assumptions with typed registries, explicit missing coverage items, and tests that fail when the modernization map drifts.

Modernization Targets

- packet types should have body schemas, compatibility entries, builder support, manifest definitions, definition parts, and runtime connector status recorded in one audit surface
- runtime mutations should map to coordinator responsibilities, prepare/finalize behavior, policy action IDs, signed corridor use, and Signal Conductor enrollment status
- Definition, Bundle, and Preference are canonical packet types with body schemas, compatibility entries, builder support, definition parts, and seed/profile coverage

- imported Definition and Bundle packets describe semantics but never introduce executable server behavior
- Trusted Runtime Coordinators and the Interface Signal Conductor should become the standard orchestration path after coverage is complete

Phase Plan

1. Save the broad plan and add coverage audits. 2. Use the audit output to prioritize packet-type definition work. 3. Expand manifest definitions and definition parts type by type, preserving existing schemas and compatibility behavior unless a type-specific schema evolution is explicitly approved. 4. Adapt runtime mutation paths into Trusted Runtime Coordinator seams behind the Interface Signal Conductor, keeping signed corridor behavior intact until each mutation has a tested replacement boundary. 5. Enroll completed types and connectors only when their docs, tests, policies, and runtime behavior are all aligned. 6. Retire planned-gap records only when the implementation and tests prove the gap is actually closed.

First Pass

The first pass is intentionally non-behavioral:

- add this chapter document
- link it from the implementation-guide index
- add packet type modernization coverage audits
- add runtime mutation modernization coverage audits
- add tests proving current types, mutation intents, Preference connector enrollment, and Definition/Bundle next-phase targets are all visible
- keep known modernization gaps green only through explicit planned-gap records

Guardrails

- historical baseline: Definition and Bundle were initially kept out of `PACKET_TYPES`; the current promotion pass enrolls them as canonical packet types
- do not change `MutationIntent`, route payloads, packet schemas, or compatibility registry behavior during the audit pass
- do not migrate legacy scope-display caches until a dedicated compatibility pass
- the initial audit baseline kept `Preference.element` out of signed mutation enrollment; the current corridor now enrolls `preference.element.set` through `prepare/finalize`
- do not rebuild generated public docs artifacts as part of this first pass

Test Plan

- run packet modernization coverage tests
- run runtime mutation modernization coverage tests
- keep packet-definition manifest and audit tests green
- keep signed route and service-writer audit tests green
- validate docs with `npm run docs:validate`

Decision Notes

The audit modules are the working checklist for the next implementation passes. Broken current invariants should fail tests. Expected gaps should remain visible as planned modernization work until they are closed by a later pass.

Trusted Runtime Coordinator Architecture

Current modernization direction is to organize secure runtime work around enrolled Trusted Runtime Coordinators rather than route-local write code.

Target coordinator families:

- Interface Signal Conductor for client-side UI signal shaping before requests leave the interface
- Trusted Request Coordinator for user/API intent routing, request normalization, and fail-closed preflight
- Trusted Definition Coordinator for active definition context, node definition preferences, definition part selection, compatibility-only definitions, and reseed readiness views
- Trusted Regulation Coordinator for policy and governance resolution
- Trusted Planning Coordinator for defaults, dependency, and bundle-plan resolution
- Trusted Building Coordinator for producing packet candidates through the generic builder pipeline
- Trusted Inspection Coordinator for structural refs, dependency satisfaction, and bundle coherence
- Trusted Testing Coordinator for schema, compatibility, and policy assertions
- Trusted Certification Coordinator for signatures, tickets, digests, and final integrity
- Trusted Archival Coordinator for packet-store writes and archive indexes
- Trusted Verification Coordinator for local packet verification and verification reports
- Trusted Import Coordinator and Trusted Export Coordinator for bundle ingress/egress

- Trusted Projection Coordinator for UI-ready graph projections and available actions

These coordinators are runtime concerns. They execute trusted local code and feed live context through the Core Contracts Vault. Packet definitions may describe allowed operations, defaults, dependencies, policy requirements, actions, and projection hints, but imported packet definitions must never execute local runtime behavior. The Interface Signal Conductor is the client-side signal former; it has no trusted authority. The Trusted Request Coordinator is the runtime front desk: it normalizes requests, preflights registered client/API intents, and hands accepted runtime requests to downstream coordinators. The Trusted Definition Coordinator is the gate for definition lookup: internal candidate listing, ranking, conflict audit, compatibility selection, and runtime-view compilation functions are routed through its public coordinator surface instead of being imported as loose helper functions. The Trusted Regulation Coordinator follows the same gated coordinator pattern for policy contexts, write-policy gates, requirement listing, and readiness audits. The Trusted Planning Coordinator owns packet operation plans, builder selection, defaults, dependencies, child-plan seams, and planning readiness so default/dependency work does not masquerade as policy enforcement.

Definition-Driven Build And Projection Direction

The same declaration language should guide both packet creation and packet projection.

Creation uses definitions, builders, defaults definitions, dependencies definitions, policy requirements, and runtime variables to produce trusted operation plans before packet candidates are built. Projection should eventually use definitions, projection hints, component keys, action keys, display fields, and graph relationships to produce safe UI-ready view models.

Resolver ownership is split by domain:

- definition resolves active definition context, definition parts, node runtime preferences, compatibility-only definitions, and runtime definition views
- planning resolves operation plans, builder selection, default packet cascades, dependency satisfaction, workflow dry-runs, and child packet/component seams
- regulation resolves policy requirements, write-policy gates, governance rules, trust gates, moderation rules, voting eligibility, and other policy checks that may be needed inside or outside packet creation
- building creates packet candidates through the generic builder pipeline
- projection resolves surfaces, display models, badges, and available actions

Builders remain packet anatomy. Defaults describe normal starting shape. Dependencies describe required structural refs. Planning assembles those pieces into a concrete operation plan and asks Regulation for the active policy envelope when policy meaning matters. Regulation does not build packets or apply defaults; it classifies policy requirements and write gates for creation, projection, import/export review, moderation, runtime reads, and governance checks. Projection definitions describe safe display and interaction hints.

Trusted Coordinator Scaffold Standard

Trusted coordinators now share a scaffold contract: public coordinator object, stable coordinator id, typed result envelope, issues, trace entries, optional request/operation ids, and a manifest entry describing expected methods. Foldered trusted coordinators expose only their public coordinator and public types from index.ts; internal function modules and registries stay private behind the coordinator surface.

`npm run audit:trusted-coordinators` checks the scaffold manifest. The audit currently treats Definition, Regulation, Planning, and Request as foldered gated coordinators. Building, Projection, and Resolution remain legacy-flat with warnings until they are promoted.

Trusted Planning Coordinator Pass

The Trusted Planning Coordinator is now the runtime middleman for operation planning. It exposes gated methods for resolving operation plans, default plans, dependency plans, builder descriptor selection, child packet plan seams, and planning readiness audits. Defaults and dependencies moved out of the Trusted Regulation Coordinator because they are construction-planning inputs, not policy enforcement by themselves.

The current implementation is intentionally pre-reseed practical:

- default plans wrap definition defaults, policy-selected default refs, and local overrides without building packet bodies directly
- dependency plans gather Definition dependency parts, workflow dependency IDs, semantic descriptors, runtime capability refs, and optional workflow dry-run findings
- builder selection picks the best runtime-ready builder descriptor for the packet type, subtype, and action IDs
- child packet plans are a typed seam, currently empty until defaults, bundles, and projection/component layout semantics are declared enough to recurse safely
- operation plans call Regulation for policy contexts and optional write-policy gates, rather than duplicating policy logic

Regulation is now policy-only. It still works outside packet creation, including projection, import/export review, moderation, runtime reads, and governance checks. Planning may call Regulation during packet creation, but Regulation does not own defaults, dependencies, or builder selection.

Manifest Core Pass

The first chunky implementation pass expanded the packet manifest from the Preference template to the active packet types with generic builder-pipeline support: Element, Location, Role, Claim, Relation, Report, Proposal, Reaction, Decision, Action, Discussion, and Policy.

This pass remains runtime-ready. The new definitions describe existing body schemas, compatibility registry posture, generic builder support, action descriptors, planner descriptors, projection/index descriptors, and Definition parts. They do not change route payloads, packet schemas, runtime mutation behavior, or Signal Conductor connector enrollment.

Builder-missing types remain explicit missing coverage items in the modernization audit. Preference later received canonical builder support and signed-corridor write enrollment.

Packet-Type Authority Pass

The next implementation pass shifted the forward-looking checklist from legacy type enrollment toward manifest packettype authority. The later promotion pass enrolled Definition and Bundle as canonical packet types while preserving packettype language as the forward-facing semantic layer.

The pass added body builders for Definition, Bundle, and Preference. Those builders now feed canonical seed-profile helpers that create real Definition packet envelopes and a Bundle packet inventory for reseed material.

Packet-type modernization coverage is now the forward-looking audit surface for manifest definitions. The legacy type coverage remains as a migration bridge for live packet types and should keep missing coverage items visible until those types are converted into packet-type definitions and runtime connectors.

Compatibility Definition Standard Pass

The compatibility standardization pass makes compatibility a required, auditable definition contract. Every manifest packet type now needs a required `packet_compatibility` Definition part and a current-version identity adapter descriptor.

Generic type definitions derive definition compatibility descriptors from the canonical compatibility registry. Current-only types expose identity compatibility, while legacy-aware types expose adjacent upcast/downcast ladder edges where the registry has adapter functions. Multi-step ladders use the fullchainbundle strategy so future bundles can carry discoverable adapter metadata without pretending every adapter must touch the current schema directly.

The manifest audit now fails when compatibility posture and descriptors disagree, when downcast edges lack loss awareness, when duplicate adapter edges exist, or when a claimed full-chain graph is disconnected from the current version. This keeps reseed and import/export planning honest before runtime handler extraction or Trusted Runtime Coordinator pipeline promotion begins.

Mutation Handler Extraction Pass

The signed mutation corridor remains the live authority for prepare, proof, finalize, and persistence decisions. The packet-runtime Signal Conductor remains the client/API-to-runtime connector bridge; it does not own signed mutation internals yet.

The extraction pass introduces domain-composed mutation handler maps for locality, discussion, reaction, assembly, relation, role, and actor policy. MutationPrepareHandlers and MutationFinalizeHandlers remain compatibility facades for the current implementation, while the composed maps give the runtime a clearer stepping-stone toward generic packet planners.

Each live mutation intent now has a genericization classification:

- generic_ready means the intent is close to manifest/action/planner routing once its local read dependencies are isolated.
- plannerextractionneeded means reusable packet planning must be extracted before generic routing.
- workflow_specific means runtime orchestration still coordinates multiple packet operations or projections.
- legacy_bridge means the intent is a compatibility alias that should collapse into a canonical intent.

This pass intentionally preserves behavior. It records which mutation-service code should be retired, which should become reusable planners, and which orchestration remains runtime-owned before any live Trusted Runtime Coordinator pipeline promotion.

Packet Operation Ontology Pass

The operation ontology pass adds the missing contract between packet definitions and trusted runtime execution. Packet definitions may now describe allowed mutation semantics by mapping their manifest mutation descriptors to known operation kinds such as `singlepacket.create`, `singlepacket.revise`, `relation.set`, `claim.assert`, `reaction.set`, `bundle.import`, `projection.refresh`, `compatibility.adapt`, and `workflow.compose`.

This ontology is an allowlist, not executable packet-defined code. Each operation records its expected planner kind, builder kind, result type, trusted local runtime engine, generic capability posture, and safety notes. Packet definitions can request known operation semantics, but only local trusted engines may execute builders, planners, adapters, workflows, or persistence.

The manifest audit now fails closed when a mutation descriptor cannot map to a known operation kind. The forward-looking packet operation modernization coverage lists every manifest mutation, the operation kinds it resolves to, the trusted local engine requested, and whether the gap is already mapped or still planned.

The signed mutation genericization audit also records operation mappings for every live mutation intent:

- `generic_ready` intents map directly to concrete operation kinds, but still wait for the later live promotion pass.
- `plannerextractionneeded` intents map to their target operation kind and keep an explicit planner extraction gap.
- `workflow_specific` intents map to `workflow.compose` or a composed operation set and remain runtime-owned until their component operations can be split safely.
- `legacy_bridge` intents point at the canonical operation direction they should collapse into.

This keeps ingress normalization separate from mutation authority. The Signal Conductor can normalize client/API ingress requests and eventually choose operation descriptors, while the trusted mutation pipeline still owns `prepare/finalize/proof/persistence` until a later pass promotes selected operation kinds through the generic path.

Generic Workflow Planner Contract Pass

The workflow planner contract pass adds the declarative layer above individual operation kinds. Packet definitions may now describe definition workflow plans as ordered steps over known generic operations, trusted resolver IDs, value bindings, simple conditions, policy action IDs, and runtime dependency IDs.

Workflow plans are data, not code. Definitions can say "resolve actor and target, then run `relation.set`" or "if the input value is present run `reaction.set`, otherwise run `reaction.clear`." They cannot introduce arbitrary functions, dynamic imports, persistence behavior, route payloads, or proof rules. The runtime interpreter validates every operation, resolver, dependency, condition operator, policy action, and step reference against local allowlists before producing a dry-run plan.

The first definition workflow plans cover the generic-ready mutation candidates:

- `relation.follow.add`
- `relation.follow.clear`
- `role_association.claim.set`
- `reaction.vote.set`

These plans do not enroll live execution. They prove the manifest can describe packet-specific variables and ordered generic work while preserving the signed mutation corridor as the only live prepare/finalize/proof/persistence authority.

Policy and dependency descriptors now matter as referenced workflow metadata, but their full semantics remain a dedicated pre-reseed pass. Unused legacy packet types remain explicit missing coverage items and do not block the generic workflow contract or switch-over planning.

Workflow Alignment Pass

The workflow alignment pass connects the manifest workflow contract to the extracted mutation planner map. It adds a runtime-side audit that lists every live mutation intent, its genericization status, operation mapping, workflow-plan coverage, policy action IDs, trusted resolver/capability IDs, and remaining packet-specific assumptions.

The alignment map is the working checklist for retiring packet-specific mutation-service code. Generic-ready intents must have clean workflow dry-runs and trusted local capability coverage. Planner-extraction intents may either have a definition workflow plan or an explicit missing coverage item. Workflow-specific intents remain runtime-owned orchestration until their component operations can be split safely. Legacy bridge intents point at canonical workflow directions rather than receiving independent workflows.

This pass expands definition workflow coverage for knowable planner-extraction candidates:

- `relation.association.add`
- `relation.association.clear`
- `relation.residence.add`
- `discussion.reply.create`

The alignment remains runtime-ready. Existing runtime planner modules are registered as trusted local capabilities by descriptor, but their implementation is not moved, rewritten, or invoked through generic execution yet. Unused removed packet types remain visible missing coverage items and do not block switch-over planning.

Runtime Ingress And Mutation Handoff Pass

Ingress preflight and mutation authority remain separate layers. Ingress preflight owns client/API normalization, manifest/workflow lookup, connector selection, definition handoff metadata, and response/refresh hints. The signed mutation corridor owns policy/proof authority, prepare/finalize lifecycle, mutation tickets, signed packet validation, persistence, and canonical mutation effects.

The handoff pass adds a definition `PacketRuntimeMutationHandoff` contract. A handoff records the normalized mutation direction, workflow alignment status, operation kinds, workflow plan IDs, trusted capability IDs, policy action IDs, dependency IDs, resolver IDs, prepare/finalize coordinator names, and return refresh hints. It carries `externaldefinitionexecution_enabled: false` to record that imported definitions describe behavior while trusted local runtime code executes it.

Generic-ready and workflow-aligned planner-extraction intents can now produce `definition_ready` handoffs. Runtime-owned workflow intents produce explicit non-ready handoffs with orchestration reason codes. Legacy bridge intents point at canonical handoff directions. Unknown mutation intents fail closed before any mutation handoff.

At the time of this pass, this did not change the live mutation routes. The current state is stricter: `NexusMutationService` remains the live signed mutation authority, and authenticated `Preference.element` writes now enter that trusted mutation service path rather than the old direct packet-runtime connector.

Packet-Based Policy, Dependency, and Client Ingress Enrollment Pass

Policy and dependency requirements are now audited as packet-backed semantics rather than a second runtime-only dependency system. Workflow plans can reference policy action IDs and dependency IDs, but those references must resolve to packet policy semantics, packet Definition dependency parts, operation ontology entries, trusted workflow resolvers, or trusted local capability metadata. Runtime descriptors may index and validate those references, but they do not define packet meaning.

Policy packets remain the semantic authority for live write-lock policy. MutationPolicyGate remains the live resolver for scope policy refs, actor security mode, proof level, and accepted proof methods. Definition workflow policy descriptors record how policy action IDs map back to that packet-based enforcement model, while manifest-only actions remain definition metadata until a later live write-policy enrollment pass.

Runtime ingress now has a client/API enrollment registry. The registry is an internal allowlist of adapter-originated transport routes and portable client intent IDs:

- /api/nexus/mutations/prepare enrolls the current signed mutation intents.
- /api/nexus/mutations/prepare enrolls the Preference.element client intent; authenticated shell preference writes now use the same prepare/finalize corridor as other claimed mutations, while /api/nexus/shell-preferences remains guest compatibility state only.
- each enrollment records route or transport source, client intent ID, mutation intent, operation kinds, workflow plans, policy actions, dependency refs, and current live mode.

Unknown or custom route/intent pairings fail ingress preflight. The preflight may resolve handoff metadata and packet-backed policy/dependency descriptors, but it does not authorize, ticket, sign, persist, finalize, or bypass the signed mutation corridor. This keeps the future generic corridor aligned with enrolled client/API ingress from web, device, automation, or other adapters instead of accepting arbitrary injected operation requests.

Interface-Neutral API Ingress Pass

The enrollment layer is interface-neutral. Web shell, Raspberry Pi controls, local automation, and future adapters should all map their local events into the same portable client intent IDs, such as scope.follow.set, scope.association.clear, discussion.reply.create, and preference.interface.set. Runtime registries should not encode web UI concepts as packet meaning.

The live API routes now consult ingress preflight before delegating to the live corridor:

- prepare parses the request intent, validates client/API ingress enrollment, then delegates to NexusMutationService;
- finalize reads the stored ticket, validates the ticket's original mutation intent against enrolled prepare ingress, then delegates to NexusMutationService;
- authenticated shell preferences use the standard prepare/finalize mutation routes with preference.element.set;
- /api/nexus/shell-preferences remains a guest compatibility route and is outside packet-runtime connector enrollment.

This pass is still not generic execution. Preflight validates allowlist and metadata alignment only; mutation policy/proof/ticketing/persistence remains authoritative.

No-Deferral Pre-Reseed Closure Program

Reseed design is now gated on full closure of in-scope live runtime modernization work. The chapter can still be split across multiple implementation passes, but the remaining live runtime work is no longer tracked as open-ended missing coverage items. The pre-reseed closure ledger classifies every live mutation intent, runtime connector path, workflow plan, policy/dependency requirement, client/API ingress enrollment, mutation handoff, and active packet type as closed, closingnow, queuedpre_reseed, or blocked.

The first proving promotion is follow relation set/clear:

- relation.follow.add
- relation.follow.clear

These intents now prepare through trusted generic workflow planning while NexusMutationService remains the signed mutation authority. API routes, route payloads, response shapes, policy action IDs, packet schemas, proof behavior, tickets, signatures, persistence, and projections remain unchanged. The promoted path uses manifest workflow metadata and trusted local relation planning to produce the same packet candidates and policy metadata as the previous mutation-service-specific follow planner path.

The remaining pre-reseed queue is explicit:

- relation, claim, and reaction generic enrollment for association, home locality, role claim, packet signal, and role reaction paths
- discussion and locality workflow decomposition for reply/thread planners, default surfaces, locality path/graph planning, and assembly creation
- packet-based policy/dependency semantic authority so Policy packets and Definition dependency parts carry enough meaning for reseed
- legacy bridge retirement for compatibility aliases that should not survive into the fresh reseed world, now closed by removing legacy bridge intents from the live prepare corridor
- a final reseed readiness audit after all in-scope modernization closure items are closed

Unused never-live packet types were pruned from active canon. They can return only through the same definition, schema, builder, seed, and audit path as any other active type.

Generic Composite Workflow Adapter Pass

Complex graph workflows now use named trusted composite adapter shapes in definition. These adapters are local runtime-owned orchestration patterns that compose known packet operations, policy actions, dependency refs, and result metadata. Packet definitions and workflow alignment may reference adapter IDs, but packet definitions still cannot provide executable code.

The first adapter shape is `composite.batch.packet_operations`, used by `locality.graph.apply`. It describes the reusable graph pattern: resolve inputs, plan structural packets, plan relation operation batches, resolve grouped policy, prepare unsigned digests, carry prepared-result metadata, and classify projection/refresh side effects as runtime return extensions.

Two additional graph-style shapes are recorded for reuse:

- `composite.defaultpacketset.ensure` for idempotent default packet-set creation, first represented by `discussion-surfaces.ensure`.
- `composite.entitycreate.withfollowups` for a primary entity packet plus optional follow-up operations, first represented by `assembly.element.create`.

This pass remains runtime-ready. Live API routes, payloads, mutation ticketing, signing, persistence, projections, and response shapes remain unchanged. Complex workflows stay queued for live promotion until adapter parity tests prove prepare/finalize behavior against the current mutation oracle.

Remaining Runtime Genericization Closure Pass

The second live generic promotion expands the trusted workflow seam beyond follow relations. The direct operation paths now enrolled behind `NexusMutationService` are:

- `relation.follow.add`
- `relation.follow.clear`
- `relation.association.add`
- `relation.association.clear`
- `relation.residence.add`
- `role_association.claim.set`
- `reaction.vote.set`

The live behavior contract is unchanged: API payloads, policy action IDs, ticketing, signatures, packet schemas, persistence, projections, and finalize handlers remain the current mutation authority. The prepare side now resolves these direct operations through trusted local generic planners for scoped Relation, role Claim, and packet-signal Reaction writes, using the current mutation planners/builders as the behavior oracle.

The remaining composed workflows now have named adapter shapes instead of open-ended gaps:

- `composite.locality_path.create.v0` for reusable entity/path creation and directory projection refresh.
- `composite.discussionthreadpost.create.v0` and `composite.discussion_reply.create.v0` for canonical Discussion(subtype: post/message) writes.

- `composite.role_reaction.set.v0` for mutual-exclusion support/dispute/clear reaction composition.
- `composite.actorwritepolicy.update.v0` for actor-owned Policy packet revision plus actor projection refresh.

Discussion follow-up is closed for the fresh canon: new top-level discussion writes use `Discussion(subtype: post)` semantics, while replies use `Discussion(subtype: message)`. `DiscussionThread`, `DiscussionPost`, and `DiscussionReply` are not active fresh packet types.

Initiative follow-up is also explicit: the fresh-reseed direction is `Action(subtype: initiative)` as the default OWA anchor for policy, template, branding, locality, voting, and governance defaults. Cause is not an active fresh packet type. This pass does not add an initiative selector or UI behavior.

Live Composite Workflow Promotion Pass

The runtime genericization lane is now closed for in-scope live prepare handling. Direct packet operations continue through the trusted generic operation seam, and composed workflows now prepare through trusted generic-composite workflow resolvers:

- `composite.locality_path.create.v0`
- `composite.locality_graph.apply.v0`
- `composite.discussion_surfaces.ensure.v0`
- `composite.assembly_element.create.v0`
- `composite.discussionthreadpost.create.v0`
- `composite.discussion_reply.create.v0`
- `composite.role_reaction.set.v0`
- `composite.actorwritepolicy.update.v0`

These resolvers execute trusted local runtime code only. Adapter descriptors describe the reusable workflow shape and audit metadata; packet definitions still cannot inject executable behavior. `MutationPrepareHandlers` remains the compatibility facade, `NexusMutationService` remains the signed mutation authority, and finalize handlers remain unchanged.

Actor write-policy update is mechanically promoted through the composite seam, and Policy packets plus Definition dependency parts are authoritative enough for the fresh genesis contract. Discussion canonicalization to top-level `Discussion(subtype: post)` plus reply `Discussion(subtype: message)` and OWA `Action(subtype: initiative)` are now part of the active reseed contract.

Initiative Action Hierarchy and Discussion Schema Readiness

The pre-reseed packet model now treats Action(subtype: initiative) as the forward initiative/work hierarchy anchor. Action packets can carry hierarchy refs plus packet-backed policy, template, and default packet-set refs so OWA defaults can be overridden without adding OWA-specific fields to Element or hardcoding defaults in runtime.

Canonical discussion shape now reserves Discussion(subtype: post) for top-level multimedia forum artifacts that start a thread, while Discussion(subtype: message) remains the reply/comment shape. Legacy thread/post/reply packet types are pruned from fresh canon.

Governance hooks remain schema-ready rather than workflow-complete: quorum, minimum trust, voter eligibility, approval thresholds, and voting gates should be expressed through packet-backed Policy/default material linked from the applicable initiative Action, scope, proposal, or definition context. Decision is the formal outcome packet; Report(subtype: decision_report) is reserved for future tally/evidence/process closure material.

Packet-Based Policy and Dependency Semantic Authority

Policy and dependency semantic authority is now closed for reseed readiness, while live governance execution remains later work. Policy current schema includes nullable defaultpolicy and governancepolicy sections. Older Policy revisions upcast those sections to explicit null; downcasts to older schema versions report loss when non-null default or governance material cannot be represented.

Policy packets are the semantic home for write locks, trust baselines, relation requirements, dependency and alignment rules, default inheritance, and governance hooks. The live write-lock path still runs through MutationPolicyGate; the new semantic helpers resolve and audit packet meaning without executing proposal/vote/decision behavior.

Definition dependencies_definition parts now carry meaningful dependency refs for packet operations, builder pipelines, action bridges, canonical packet-type builders, Preference projections, Bundle inventory building, and trusted compatibility/projection seams. Workflow and runtime dependency IDs must resolve through one of these anchors, Policy packet semantics, the operation ontology, a trusted workflow resolver, or an explicit trusted local engine contract.

The seeded OWA Action(subtype: initiative) now links to default-inheritance and governance-baseline policies. Forward default/policy resolution uses the Action initiative anchor.

Canonical Subtype Reset

The pre-reseed reset prunes inactive and legacy packet types from active canon. Fresh canon now includes only Definition, Bundle, Element, Location, Role, Claim, Relation, Report, Proposal, Reaction, Decision, Action, Discussion, Policy, and Preference.

Every active packet body uses top-level body.subtype as its packet classifier. Fresh writes reject old top-level classifier names such as kind, policykind, rolekind, proposalkind, claimkind, and reaction_value. Nested rule mechanics can still use precise names such as quorum or threshold kind when they are not packet classifiers.

Cause, Signal, separate initiative/work types, separate discussion thread/post/reply/forum/space types, Minutes, Artifact, and other pruned types are not valid fresh packet types. The alpha database is expected to be archived and wiped rather than adapted into fresh canon.

Final Pre-Reseed Wrap-Up

The final wrap-up retires the remaining live legacy bridge mutation intents from fresh writes:

- association.claim.set
- residence.claim.set

Canonical writes now enter through relation.association.add, relation.association.clear, and relation.residence.add. Historical legacy claim material remains readable/importable/projectable through compatibility surfaces, but the signed mutation prepare corridor, client ingress registry, handoff coverage, and live write-policy action list no longer enroll the legacy bridge intents.

The final readiness handoff lives in runtime audit code as createFinalPreReseedReadinessReport(). It records canonical write intents, compatibility-only legacy surfaces, OWA seed/default anchors, required default policies, discussion default packets, canonical definition packet types, and out-of-scope never-live packet types. Reseed design should start from that report rather than rediscovering chapter state from scattered modernization audits.

Definition Packetization and Preference Fortress Promotion

Active manifest definitions now have canonical packet material. `buildDefinitionPacketSeedEnvelopes()` emits schema-validated Definition packet envelopes for every active manifest definition part, and `buildDefinitionBundleSeedEnvelope()` groups those envelopes into one `Bundle.packet_set` definition profile inventory. `auditSeededPacketDefinitionProfile()` compares that packet material back to the core manifest and fails on missing parts, duplicate or stale bundle refs, digest drift, or manifest/profile mismatch.

Definition, Bundle, and Preference are now first-class canonical packet types. The active definition profile is seeded as real packet material, but execution remains trusted-local: imported Definition or Bundle packets can describe schemas, operations, policies, dependencies, planners, and builders, but cannot introduce executable server behavior.

This is packetized seed truth, not imported-code execution. Stored Definition and Bundle packets may describe schemas, operations, policies, dependencies, planners, and builders; trusted local runtime registries remain the only executable authority.

Claimed `Preference.element` writes now use the signed mutation `prepare/finalize` path as `preference.element.set`. The client prepares through `/api/nexus/mutations/prepare`, signs the prepared Preference packet candidate, finalizes through `/api/nexus/mutations/finalize`, and then receives the same projected Preference result shape from the mutation result. `/api/nexus/shell-preferences` is now guest compatibility state only. The old direct `preference.element.interface.set` connector is retained as a definition/internal comparison bridge rather than the live claimed-write path.

Reaction packet convergence pass

Current pre-reseed canon collapses lightweight votes, packet signals, support/dispute posture, and emoji-style emotional responses into Reaction as the single packet type for target-agnostic responses.

Reaction packets are replaceable per actor/target/context. A revision can carry any combination of:

- `vote_value`: 'up', 'down', or null
- `attestation_value`: support, dispute, or null
- `emoji_keys`: a bounded list of basic emoji keys

Reaction does not encode target packet type or target-specific purpose. Proposal voting, discussion up/down signaling, role support/dispute posture, and later emoji/reaction UI should all route through the same packet type and projection layer. The old standalone Vote and Attestation packet types are removed from fresh canon for the clean pre-reseed path.

Trusted Resolution Coordinator and Projection Definition Pass

The trusted runtime coordinator family now has a shared home under `runtime/trusted_coordinators/*`. Direct generic workflow promotion, composite workflow promotion, composite adapters, resolution, and projection share this layer instead of living as one-off files under the Nexus server adapter folder.

A portable resolution DSL now lives in core as declaration language, not executable behavior. It defines shared binding shapes, resolution steps, and preset descriptors such as primitive bindings, packet refs, policy gates, dependency gates, relation lookup, discussion thread context, role scope context, compatibility projection, and UI card projection. Packet definitions and workflow descriptors may point at preset IDs, while trusted runtime coordinators decide how those presets are actually resolved locally.

Packet projection descriptors are now richer definition data. They can declare field bindings, layout/component keys, preferred surfaces, action registry keys, dependency IDs, policy action IDs, and resolver preset IDs. The generic packet definitions now seed default summary-card and detail-panel projections for each active packet type, giving the UI a definition-driven shape without embedding UI-specific packet law into core logic.

Definition part body building now carries descriptor payloads for action registries, builders, planners, workflow plans, projections, compatibility adapters, and dependency descriptors. This makes Definition packets more useful for reseed instead of merely listing IDs.

This pass is still conservative at the product edge. The shared packet action service now asks projection definitions for the preferred packet surface, but route-level UI layouts are not yet fully generated from projection descriptors. The next useful pass is to wire projection view models into universal packet card/detail components and reseed Definition packets with these richer descriptor bodies.

Architecture And Layers

Architecture shape

The long-term architecture is a three-layer stack with strict boundaries:

Core Nexus / Core Contracts Vault

The portable packet law and machinery.

- packet schemas, definitions, builders, compatibility, and validation
- defaults definitions and dependencies definitions

- policy requirement contracts and deterministic graph logic
- canonicalization, signing, verification, import/export/merge rules
- no database ownership, private-key custody, route assumptions, UI state, or platform-specific behavior

Runtime / Trusted Runtime Coordinators

The trusted execution environment around Core.

- session and actor context
- request dispatch and validation
- policy/regulation resolution against live packet state
- packet planning, building, inspection, testing, certification, archival, verification, import, export, and projection
- storage adapters and API-facing glue
- signing-ticket flow and mutation finalization

Runtime coordinators feed live context through the Core Contracts Vault and execute trusted local code. Packet definitions can describe allowed behavior, but imported definitions do not execute runtime behavior.

Interface / Signal Cockpit

The reusable user-facing cockpit.

- surfaces, page layout builders, cards, menus, tabs, badges, forms, modals, and loading overlays
- lightweight input validation and intent emission
- ticket/signing prompts, confirmations, errors, success states, and refresh requests
- projection consumption through view models

The interface should not decide packet validity, policy authority, or graph truth. It submits user intent and renders trusted projections.

The load-bearing rule stays the same: data is the system; platforms are adapters.

Product framing

The current product stack remains:

- FCF = principles and coordination physics
- Nexus = portable signed-packet substrate and graph engine
- OWA = geography-first civic application profile on Nexus

OWA may provide default policies, default packets, projection preferences, governance presets, and seed choices. OWA should not mutate Nexus packet builders or redefine core packet anatomy.

Repository boundaries

The active repo split is:

- core/* for portable packet logic, schemas, builders, interpreters, definition contracts, pure projections, and graph rules
- runtime/* for trusted coordinators, persistence, runtime services, query services, auth/trust/discussion orchestration, and API glue
- app/* for Signal Cockpit components, hooks, constants, public content, reusable UI, and shared state
- src/app/* for the Expo Router route shell and API entrypoints

Documentation system

The multi-chapter internal docs now use a chapter-first source-of-truth model.

Canonical content lives in:

- docs/implementation-guide/*
- docs/specifications/*
- docs/roadmap/*

The top-level files:

- docs/implementation-guide.md
- docs/specifications.md
- docs/roadmap.md

remain short local index shells only.

Public docs generation is now expected to follow that structure:

- docs/public/public-docs.manifest.json owns the ordered public source lists
- scripts/validate-public-docs.mjs validates manifest and shell/source-of-truth rules
- scripts/build-public-docs.mjs compiles readable docs data, Markdown downloads, PDF downloads, and version records
- generated artifacts under app/public/generated/, public/downloads/, and docs/public/version-records/ are derived outputs and should not be edited by hand

Expected workflow:

- update chapter files first

- run npm run docs:validate after canon doc edits
- rely on npm run docs:build during site/export builds or when intentionally refreshing generated public docs artifacts

Core versus adapters

Core owns

- packet schemas and validation
- packet definitions, builders, defaults definitions, and dependencies definitions
- graph relationships and traversal
- import/export/merge rules
- signing and verification law
- policy requirement hooks and deterministic business logic

Runtime owns

- trusted coordinator workflows
- live packet lookup and storage writes
- actor/session context
- signing-ticket flow
- policy resolution against current state
- verification reports and runtime-owned indexes
- API-facing orchestration

Adapters and interface own

- routing and navigation
- rendering and layout
- input UX
- loading, confirmation, and error chrome
- device or browser integration
- projection display

Adapters should not invent parallel business logic for trust, validation, compatibility, policy, or merge behavior.

Nexus scoped loading

Nexus loading chrome is an app-layer concern. The scoped loading provider, boundary, overlay, and hook live under app/components/nexus/ui/feedback/loading/* and track caller-owned visual scopes such as a page, panel, tab, card, menu, or action pane.

The loading system must not depend on packet type, mutation intent, packet action registry entries, or runtime operation categories. Runtime and core code should remain unaware of blur overlays, spinners, and UI input blocking.

The active state starts immediately so duplicate input inside the same boundary can be blocked before the delayed visual overlay appears. The visible state is delayed to prevent spinner flicker on fast operations, and visible overlays may remain briefly to satisfy the minimum visible duration.

Shared Nexus buttons and action-menu items can carry optional loading scopes, allowing common action chrome to start scoped loading without knowing packet types or runtime operation semantics. The UI site still owns the visual scope choice because it knows which page, panel, tab, card, menu, or action pane should be blocked.

Navigation and shell model

The intended navigation model still uses three mental axes:

- place or scope
- topic or work
- time or evidence

The current implemented Nexus slice is a practical early version of that model, with:

- dashboard
- discussions
- votes
- roles
- trust
- library
- packet explorer as a shell-level inspection workspace
- account and identity custody as wrapper-level surfaces

Nexus UI Component Catalog

Purpose

This catalog records the current Nexus UI component landscape before a broader universal-template cleanup. It is an audit artifact only: it should guide later consolidation work without moving files, changing visuals, or coupling Nexus UI to the public site design system.

Public-site components may be useful references, but Nexus should keep its own component system because the public site and Nexus workspace are separate product surfaces.

For exact file-level counts, route-local component candidates, raw React Native primitive usage, and extraction sequencing, see Nexus UI Hard Inventory.

Current pressure points

- `src/app/nexus/locality/create.tsx` and `src/app/nexus/discussions.tsx` are the largest route-local UI surfaces and contain many one-off controls, pickers, composers, modals, and layout sections.
- `app/components/nexus/nexus-ui.tsx` is now a temporary compatibility bridge; shared primitive implementations live in focused `app/components/nexus/ui/*` component-type modules.
- `app/components/nexus/ui/tabs/nexus-tabs.tsx`, `app/components/nexus/ui/tabs/nexus-tab-primitives.tsx`, and `app/components/nexus/packet-explorer/nexus-packet-explorer-tab-deck.tsx` already share some tab skeleton pieces, but the Explorer tab deck remains a separate document-tab system.
- Modal and overlay behavior is spread across route-local Modal usage, auth gates, feature-status explainers, packet explorer overlays, sidebar drawers, and loading boundaries.
- Action chrome is the most mature shared area: packet cards, badge strips, menu buttons, action menus, action lists, and scoped loading plumbing already form a reusable foundation.

Folder foundation status

The first Nexus UI folder foundation pass moved stable shared modules into `app/components/nexus/ui/*` without changing visuals or route structure:

- `loading/` now lives at `ui/feedback/loading/`
- `action-card/` now lives at `ui/cards/action-card/`
- `action-list/` now lives at `ui/actions/action-list/`
- `nexus-tabs.tsx` and `nexus-tab-primitives.tsx` now live under `ui/tabs/*`

The older locations should not be reused for new shared UI. Larger feature folders and route-local surfaces are intentionally unchanged until later extraction passes.

Overlay consolidation status

The first universal-template consolidation pass added shared overlay primitives under `app/components/nexus/ui/overlays/*`:

- NexusModalShell owns the reusable Modal backdrop, close press target, centering, and Nexus card frame.
- NexusOutcomeDialog and NexusConfirmDialog compose common outcome and confirmation dialog patterns on top of the shell.
- NexusPopover reserves a lightweight shared frame for caller-positioned overlay panels.

The initial migration moved repeated modal chrome in dashboard packet validation, packet Explorer import outcomes, and locality create/reuse/picker dialogs onto the shared shell while preserving their existing visual classes and handlers. Auth/session logic remains outside the generic overlay primitives.

Primitive split status

The broad nexus-ui.tsx primitive file has been split into focused component-type modules while preserving visual behavior:

- card surfaces live in ui/cards/nexus-card.tsx
- action buttons and chevrons live in ui/actions/*
- badges live in ui/feedback/nexus-badge.tsx
- segmented controls and inline selects live in ui/forms/*
- chrome, appearance, bevel, and section-header helpers live in ui/layout/*
- attached tab rails live in ui/tabs/nexus-attached-tab-rail.tsx

Nexus callers should import from @app/components/nexus/ui or a direct family path. app/components/nexus/nexus-ui.tsx remains only as a temporary bridge for compatibility and should not be the source for new imports.

Feature extraction status

The first large route decomposition passes created app/components/nexus/features/discussions/* for discussion-specific composed UI. The route still owns query normalization, data loading, mutations, auth gates, router navigation, and reply branch state, while extracted feature components render feed/thread/post workspace panels, feed post cards, root post cards, reply trees, vote/reply-count pills, and post/reply composers.

Discussion loading scopes are visual-boundary owned: feed load-more, root replies, reply branches, vote pills, and post/reply composers now have scoped loading seams without coupling loading to packet action registries or runtime mutation types.

The workspace panel shapes are intentionally feature-local but generic-friendly: feed panels, thread toolbars, recursive tree sections, and composer panels can be promoted into ui/* later if another Nexus surface proves the same reusable skeleton.

The sidebar has also started feature-local extraction under `app/components/nexus/features/sidebar/*`. `nexus-sidebar.tsx` remains the shell controller for router, auth, preference, rail-state, and scope mutation handling, while rail toggles, preference switches, function menus, scope rows, grouped scope sections, and scope action menus now live in the sidebar feature module.

Locality create has begun feature-local extraction under `app/components/nexus/features/locality/*`. The route still owns params, identity/auth gates, draft persistence, graph/path state, API calls, mutation handlers, and modal state, while search/build panels, picker dialogs, outcome dialogs, graph rows, and preview panels now live in locality feature components. The obsolete level-row ladder builder path was removed after extraction; the active create flow is graph-based.

Component Type Catalog

| Component type | Current files and usages | Current status | Duplication notes | Future template target | Migration risk | | --- | --- | --- | --- | --- | --- | | Cards and packet cards | `ui/cards/nexus-card.tsx`, `ui/cards/action-card/`, `focus/`, `dashboard/`, `route-local` nested cards in `trust`, `roles`, `votes`, `library`, `discussions`, `locality/create`, and packet explorer panels | Shared base plus feature-specific compositions | NexusCard is used widely, but route files still compose repeated metric cards, inset cards, warning cards, focused packet panels, and packet preview rows manually | `ui/cards` with base surface, metric/stat card, packet card, focused packet panel, warning/outcome card, and repeated inset section patterns | Medium | | Action buttons, menus, lists, and badge strips | NexusActionButton and NexusChevronIcon in `ui/actions/`; `ui/cards/action-card/`; `ui/actions/action-list/`; packet action conversion in `packet-actions/`; `dashboard queue/list` actions | Strongest shared foundation | Shared action menus and badge strips exist, but route-local action clusters and raw buttons still appear in large pages; button/menu scoped loading is now available but adoption is caller-owned | `ui/actions` with action button, icon/menu button, action menu, action cluster, action list, badge strip, and packet-action adapters | Low | | Tabs and tab decks | `ui/tabs/nexus-tabs.tsx`, `ui/tabs/nexus-tab-primitives.tsx`, route uses in `roles`, `identity/sign-in`, `locality/create`, packet explorer toolbar and primary rail; Explorer document tabs in `packet-explorer/nexus-packet-explorer-tab-deck.tsx` | Shared navigation tabs plus separate Explorer document tabs | Function-surface tabs and Explorer tabs both use NexusTabFrame / NexusTabLabel, but differ in close controls, tooltip behavior, wrapping, and session/document semantics | `ui/tabs` with shared frame/label/close primitives, navigation rail/stack, segmented tabs, and Explorer document-tab extension | Medium | | Modals, gates, confirmations, and overlays | `ui/overlays/`, `features/locality/`, `nexus-auth-gate.tsx`, `nexus-shell-entry-gate.tsx`, `nexus-feature-status-context.tsx`, packet explorer shell overlay, remaining route-local overlay variants | Shared modal shell now exists; gates remain specialized | Dashboard validation, packet import outcome, and locality create/reuse/picker dialogs now use shared modal chrome; locality dialog content is feature-local while auth/entry gates and feature-status overlays still need careful migration because they carry session or shell-specific behavior | `ui/overlays` with modal shell, confirmation dialog, outcome dialog, anchored popover, blocking gate, and shell overlay host | Medium | | Dropdowns, selects, pickers, and menus | NexusInlineSelect in `ui/forms/`; `ui/forms/search/`; NexusActionMenu; locality kind/parent/result pickers; identity location search; packet explorer search/export lookup lists; sidebar menus | Mixed shared and route-local | Action menus and inline selects are shared; search dropdown/result chrome now has shared field/list/row primitives, while picker-specific side effects remain local | `ui/menus` or `ui/forms` with inline select, anchored menu, searchable result list, picker modal, and selectable row | Medium | | Text inputs, composers, search fields, and form rows | `ui/forms/nexus-field-shell.tsx`, `ui/forms/nexus-text-input.tsx`, `ui/forms/nexus-text-area.tsx`, `ui/forms/nexus-field-action-row.tsx`, `ui/forms/search/`, `features/discussions/`; raw TextInput remains in larger route-local specialized controls | Shared field/input/search chrome is now available | Identity fields now wrap the generic shell/input primitives; Packet Explorer import/export fields, trust notes, roles support comments, and discussion post/reply composers use shared input/textarea seams. Broader route-local form sections still need careful migration | `ui/forms` with field shell, text input, textarea/composer, search box, result list, error text, hint text, and field action row | Medium | | Loading and feedback states | `ui/feedback/loading/`; `ui/feedback/nexus-feedback-states.tsx`; discussion and locality feature loading scopes; local isLoading flags in route pages; warnings/errors with NexusCard tones | Shared loading and basic feedback cards now exist | Scoped loading provider/boundary exists; discussions mount visual-boundary scopes for

Route-local areas to audit first during consolidation

1. `src/app/nexus/locality/create.tsx`

- Remains the controller for params, auth gates, draft state, graph/path derivation, API calls, mutations, and handler ownership.
- First feature extraction now lives in `features/locality/*`: search/build panels, picker dialogs, selected-result/remove/outcome dialogs, graph rows, preview panels, and shared locality-create types. The legacy level-row builder was removed; remaining cleanup should target repeated picker/list skeletons only after another surface proves reuse.

2. `src/app/nexus/discussions.tsx`

- Still owns feed/thread/post workspace state, routing, loading, mutations, auth gates, reply branch state, and pagination actions.
- Feature extraction now lives in `features/discussions/*`: feed/thread/post panels, feed cards, root post cards, vote/reply-count pills, reply tree controls, and post/reply composers. Future work can consider promoting workspace panel shells, scrollable feed panels, thread toolbars, composer shells, reaction/vote skeletons, and recursive tree rails only after non-discussion reuse appears.

3. `app/components/nexus/nexus-sidebar.tsx`

- Still owns shell state, router/auth hooks, preference persistence, rail animation, and scope mutation handlers.
- First feature extraction now lives in `features/sidebar/*`: rail toggles, guest avatar, preference switch, current context card, function menu rows, scope section headers, grouped scope rows, scope action menus, and scope menu content. Future work can promote rail toggles, compact nav rows, grouped collapsible sections, and anchored compact action menus only after other surfaces need the same shape.

4. `app/components/nexus/packet-explorer/*`

- Contains Explorer-specific tabs, toolbars, import/export/search panels, data panels, link groups, and modal outcomes.
- Best first extraction targets: document-tab extension, workbench panel frame, lookup/result list, import/export outcome dialog.

5. Identity routes and `nexus-identity-ui.tsx`

- Already have identity field primitives, but route-local selection cards, search results, passphrase forms, and security controls overlap with broader Nexus form needs.
- Best first extraction targets: field shell generalization, selectable row/card, security preference row.

Recommended target organization

Future consolidation should create a Nexus-native shared UI home without moving everything at once:

- app/components/nexus/ui/actions
- action buttons, icon/menu buttons, action menus, action clusters, action lists, badge strips
- app/components/nexus/ui/cards
- base cards, packet cards, metric/stat cards, warning/outcome cards, focus/preview cards
- app/components/nexus/ui/tabs
- tab primitives, navigation rails/stacks, segmented controls, Explorer document-tab extensions
- app/components/nexus/ui/overlays
- modal shell, confirmation dialog, outcome dialog, popover, feature-status overlay, auth/entry gates
- app/components/nexus/ui/forms
- field shell, text input, textarea/composer, action row, search/*, picker shell, toggles
- app/components/nexus/ui/layout
- page frame, scroll frame, section header, panel/workbench panel, section band, toolbar row, metric grid, rail section, panel split, responsive content frame
- app/components/nexus/ui/feedback
- loading boundary/overlay exports, empty state, error state, warning state, status rows

The existing folders can then become either adapters around the shared UI or feature-specific compositions:

- packet-explorer/* should keep Explorer state/workbench logic but use shared tabs, overlays, forms, feedback, and panel frames.
- features/locality/* keeps locality graph/search semantics but uses shared picker, form/search, modal, loading, and warning/outcome components.
- features/discussions/* keeps discussion-specific workspace, post/reply/vote composition while consuming shared UI primitives and caller-owned loading scopes.
- features/sidebar/* keeps sidebar-specific rail, preference, function-menu, scope-list, and scope-action composition while consuming shared UI primitives and caller-owned loading scopes for async scope actions.
- ui/overlays/, ui/cards/action-card/, ui/actions/action-list/, ui/feedback/loading/, and ui/tabs/* now form the initial shared UI foundation.
- ui/forms/search/* now provides generic search field, result-list, result-row, status, empty/error, and loading-boundary wrappers. It intentionally does not own packet, identity, or locality search behavior.

- ui/forms/* now also provides generic field shell, text input, text area, and field action row primitives. Identity-specific field wrappers remain as compatibility adapters around those generic forms.
- ui/feedback/* now provides basic inline loading plus empty, error, warning, status, and operation card primitives. Scoped loading remains visual-scope owned by callers.
- ui/layout/* now provides page/scroll frames, panel/workbench wrappers, panel headers, section bands, toolbar rows, and metric grids. Panel loading remains caller-owned by visual scope.

Migration guidance

- Do not begin with a mass folder move. Promote one repeated pattern at a time.
- Preserve live visuals exactly unless a later task explicitly asks for design changes.
- Prefer extracting route-local UI only after at least two call sites need the same behavior.
- Keep runtime/core untouched; this is presentation infrastructure only.
- Treat public-site components as reference material, not dependencies.
- For each extraction pass, update this catalog or replace it with a more precise design-system chapter once the target folder structure becomes real.

Nexus UI Hard Inventory

Purpose

This chapter is the hard inventory behind the higher-level Nexus UI component catalog. It is a snapshot of the current Nexus UI landscape used to plan universal template extraction without changing live visuals or route behavior.

This is an audit artifact only. It should guide later consolidation, not imply that every item listed should move immediately.

Post-inventory folder foundation

After this snapshot, the first safe folder foundation pass moved stable shared modules into the canonical app/components/nexus/ui/* tree without visual or route behavior changes:

- app/components/nexus/loading/ -> app/components/nexus/ui/feedback/loading/
- app/components/nexus/action-card/ -> app/components/nexus/ui/cards/action-card/
- app/components/nexus/action-list/ -> app/components/nexus/ui/actions/action-list/
- app/components/nexus/nexus-tab-primitives.tsx ->
app/components/nexus/ui/tabs/nexus-tab-primitives.tsx

- app/components/nexus/nexus-tabs.tsx -> app/components/nexus/ui/tabs/nexus-tabs.tsx

The first consolidation pass after the folder foundation added app/components/nexus/ui/overlays/* and moved repeated modal chrome in dashboard validation, packet Explorer import outcomes, and locality create/reuse/picker dialogs onto NexusModalShell while preserving existing visual classes and handlers.

The broad app/components/nexus/nexus-ui.tsx primitive file has since been split into focused ui/actions, ui/cards, ui/feedback, ui/forms, ui/layout, and ui/tabs modules. nexus-ui.tsx remains as a compatibility bridge only; new shared UI imports should use @app/components/nexus/ui or direct family paths.

The discussions route has since started its feature extraction into app/components/nexus/features/discussions/*. Feed/thread/post panels, feed/root post cards, vote/reply-count pills, recursive reply tree controls, and post/reply composers now live there, while src/app/nexus/discussions.tsx remains the route controller for query state, loading, mutations, auth gates, and reply branch state.

The sidebar has since started its feature extraction into app/components/nexus/features/sidebar/*. Rail toggles, guest avatar, preference switch, current context card, function menu content, scope section headers, grouped scope rows, scope action menus, and scope menu content now live there, while app/components/nexus/nexus-sidebar.tsx remains the shell controller for routing, auth gates, preference persistence, rail animation, and scope mutation handling.

The counts below remain useful as the pre-move hard inventory. New shared UI should prefer the ui/* paths.

Snapshot summary

- Scope inspected: app/components/nexus/ and src/app/nexus/.
- Files inspected: 83 non-test TypeScript/TSX files.
- Total lines inspected: 29662.
- Zones: feature 30, mixed 9, route 15, shared 19, support 10.
- Raw React Native primitive references: Pressable 158, ScrollView 62, TextInput 32, Modal 30, Switch 3, ActivityIndicator 2.
- Shared Nexus primitive references: NexusCard 358, NexusBadge 201, NexusActionButton 187, NexusSectionHeader 19, NexusSegmentedPill 17, NexusPreviewPanel 14, NexusTabFrame 12, NexusTabRail 12, NexusTabLabel 8, NexusActionList 7, NexusActionListItem 5, NexusInlineSelect 5, NexusActionMenu 4, NexusFocusedPacketSection 4, NexusTabStack 4, NexusActionCard 2, NexusLoadingBoundary 2.

Largest UI pressure files

| Lines | Zone | File | Primary component candidates | Raw primitives | | ---: | --- | --- |
 --- | --- | | 2470 after feature extraction and cleanup | route |
 src/app/nexus/locality/create.tsx | NexusLocalityCreatePage controller | Pressable 0, direct
 modal chrome moved to features/locality/*, TextInput refs retained, ScrollView 4 | | 2447 |
 route | src/app/nexus/discussions.tsx | InlineReplyComposer, DiscussionVotePill,
 ReplyCountPill, ReplyNode, ReplyTree, NexusDiscussionsPage | Pressable 9, TextInput 4,
 ScrollView 7 | | 2056 | support | app/components/nexus/identity-shell-context.tsx |
 IdentityShellContext, IdentityShellProvider | - | | 1962 | mixed |
 app/components/nexus/nexus-sidebar.tsx | NexusGuestAvatar, NexusRailToggle,
 NexusPreferenceSwitch, NexusCurrentContextCard, NexusMenuSectionLabel, NexusPrimaryNavItem |
 Pressable 26, ScrollView 6, Switch 2 | | 1261 | shared | app/components/nexus/nexus-ui.tsx |
 NexusThemedBevelEdges, NexusBevelEdges, NexusCard, NexusSectionHeader, NexusBadge,
 NexusChevronIcon | Pressable 18, Modal 3, ScrollView 3 | | 967 | feature |
 app/components/nexus/packet-explorer/nexus-packet-explorer-import-panel.tsx |
 ImportResultCard, ImportHistoryCard, NexusPacketExplorerImportPanel | Pressable 2, Modal 3,
 TextInput 2 | | 859 | support | app/components/nexus/nexus-shell-context.tsx |
 NexusShellContext, NexusShellProvider | - | | 795 | feature |
 app/components/nexus/packet-explorer/nexus-packet-explorer-export-panel.tsx |
 ExportPreviewCard, NexusPacketExplorerExportPanel | Pressable 3, TextInput 6 | | 772 | mixed
 | app/components/nexus/nexus-packet-explorer.tsx | NexusPacketExplorer | Pressable 9, Modal
 3 | | 768 | feature | app/components/nexus/packet-explorer/nexus-packet-explorer-content.tsx
 | NexusPacketExplorerSeededSummary, NexusPacketExplorerLineagePanel,
 NexusPacketExplorerActionsPanel, NexusPacketExplorerValidationPanel,
 NexusPacketExplorerContent | ScrollView 3 | | 714 | route | src/app/nexus/roles.tsx |
 NexusRolesPage | TextInput 2, ScrollView 3 | | 676 | shared |
 app/components/nexus/nexus-tabs.tsx | NexusTabButton, NexusTabRail, NexusTabStack |
 ScrollView 5 | | 659 | route | src/app/nexus/trust.tsx | NexusTrustPage | TextInput 2,
 ScrollView 3 | | 655 | route | src/app/nexus/dashboard.tsx | NexusDashboardPage | Pressable
 2, Modal 3, ScrollView 3 | | 639 | route | src/app/nexus/identity/sign-in.tsx |
 NexusIdentitySignInPage | Pressable 3 |

Raw primitive inventory

These are the files most likely to contain local button, modal, form, scroll, or overlay behavior that may eventually be promoted into Nexus-native templates. Counts are lexical references, not rendered instance counts.

| File | Pressable | Modal | TextInput | ScrollView | Other raw primitives | | --- | ---: |
 ---: | ---: | ---: | --- | | src/app/nexus/locality/create.tsx | 0 after feature extraction
 cleanup | 0 direct modal shells | 3 | 4 | legacy level-row builder removed | |
 app/components/nexus/nexus-sidebar.tsx | 26 | 0 | 0 | 6 | Switch 2 | |
 app/components/nexus/nexus-ui.tsx | 18 | 3 | 0 | 3 | - | | src/app/nexus/discussions.tsx | 9
 | 0 | 4 | 7 | - | | app/components/nexus/nexus-packet-explorer.tsx | 9 | 3 | 0 | 0 | - | |
 app/components/nexus/features/locality/locality-create-graph-row.tsx | 9 | 0 | 3 | 0 | - | |
 app/components/nexus/action-card/nexus-card-badge-strip.tsx | 8 | 3 | 0 | 0 | - | |
 app/components/nexus/packet-explorer/nexus-packet-explorer-export-panel.tsx | 3 | 0 | 6 | 0
 | - | | app/components/nexus/nexus-identity-ui.tsx | 3 | 0 | 3 | 3 | - | |
 src/app/nexus/dashboard.tsx | 2 | 3 | 0 | 3 | - | |
 app/components/nexus/packet-explorer/nexus-packet-explorer-import-panel.tsx | 2 | 3 | 2 | 0
 | - | | app/components/nexus/nexus-tab-primitives.tsx | 7 | 0 | 0 | 0 | - | |
 app/components/nexus/preview/nexus-preview-panel.tsx | 3 | 0 | 0 | 3 | - | |
 src/app/nexus/roles.tsx | 0 | 0 | 2 | 3 | - | | app/components/nexus/nexus-tabs.tsx | 0 | 0
 | 0 | 5 | - | | src/app/nexus/trust.tsx | 0 | 0 | 2 | 3 | - | |
 app/components/nexus/nexus-auth-gate.tsx | 4 | 0 | 0 | 0 | - | | src/app/nexus/library.tsx |
 0 | 0 | 0 | 4 | - | | src/app/nexus/account.tsx | 0 | 0 | 0 | 3 | Switch 1 | |
 app/components/nexus/packet-explorer/nexus-packet-explorer-content.tsx | 0 | 0 | 0 | 3 | - |
 | src/app/nexus/identity/sign-in.tsx | 3 | 0 | 0 | 0 | - | |
 app/components/nexus/features/locality/locality-create-preview-panel.tsx | 3 | 0 | 0 | 0 | -
 | | app/components/nexus/packet-explorer/nexus-packet-explorer-tab-deck.tsx | 0 | 0 | 0 | 3
 | - | | src/app/nexus/votes.tsx | 0 | 0 | 0 | 3 | - | |
 app/components/nexus/action-card/nexus-action-menu.tsx | 3 | 0 | 0 | 0 | - | |
 app/components/nexus/discussions/nexus-discussion-focus-panel.tsx | 3 | 0 | 0 | 0 | - | |
 app/components/nexus/action-list/nexus-action-list-item.tsx | 3 | 0 | 0 | 0 | - | |
 app/components/nexus/focus/nexus-focused-packet-section.tsx | 3 | 0 | 0 | 0 | - | |
 app/components/nexus/action-card/nexus-card-menu-button.tsx | 3 | 0 | 0 | 0 | - | |
 app/components/nexus/packet-explorer/nexus-packet-explorer-search-panel.tsx | 0 | 0 | 2 | 0
 | - | | app/components/nexus/nexus-shell.tsx | 2 | 0 | 0 | 0 | - | |
 app/components/nexus/nexus-feature-status-context.tsx | 2 | 0 | 0 | 0 | - | |
 app/components/nexus/nexus-shell-entry-gate.tsx | 2 | 0 | 0 | 0 | - | |
 app/components/nexus/loading/nexus-loading-overlay.tsx | 0 | 0 | 0 | 0 | ActivityIndicator 2
 | | app/components/nexus/loading/nexus-loading-boundary.tsx | 2 | 0 | 0 | 0 | - |

Shared component inventory

These files already behave like shared UI substrate or close relatives. They should generally be reorganized before route files are rewritten.

| File | Lines | Exported/public components | Local component helpers | Families | Shared
 primitives referenced | | --- | ---: | --- | --- | --- | --- | |
 app/components/nexus/action-card/nexus-action-card.tsx | 81 | NexusActionCard | - | actions,
 cards | NexusActionCard 1, NexusCard 4 | |
 app/components/nexus/action-card/nexus-action-menu-controller.tsx | 159 |
 NexusActionMenuControllerProvider | NexusActionMenuControllerContext | actions, cards |
 NexusActionMenu 2 | | app/components/nexus/action-card/nexus-action-menu.tsx | 139 |
 NexusActionMenu | - | actions, cards | NexusActionMenu 1 | |
 app/components/nexus/action-card/nexus-card-action-cluster.tsx | 137 |
 NexusCardActionCluster | - | actions, cards | - | |
 app/components/nexus/action-card/nexus-card-badge-strip.tsx | 484 | NexusCardBadgeStrip |
 NexusCardBadgeIconView, NexusCardBadgeButton | actions, cards, overlays | - | |
 app/components/nexus/action-card/nexus-card-menu-button.tsx | 57 | NexusCardMenuButton | - |
 actions, cards | - | | app/components/nexus/action-list/nexus-action-list-item.tsx | 108 |
 NexusActionListItem | - | actions | NexusActionListItem 1 | |
 app/components/nexus/action-list/nexus-action-list.tsx | 40 | NexusActionList | - | actions
 | NexusActionList 1 | | app/components/nexus/loading/nexus-loading-boundary.tsx | 52 |
 NexusLoadingBoundary | - | feedback | NexusLoadingBoundary 1 | |
 app/components/nexus/loading/nexus-loading-context.tsx | 341 | NexusLoadingProvider |
 NexusLoadingContext | cards, feedback | - | |
 app/components/nexus/loading/nexus-loading-overlay.tsx | 56 | NexusLoadingOverlay | - |
 overlays, feedback | - | | app/components/nexus/nexus-tab-primitives.tsx | 427 |
 NexusTabFrame, NexusTabLabel, NexusTabDetail, NexusTabCloseButton | NexusTabBevelEdges |
 actions, tabs | NexusTabFrame 1, NexusTabLabel 1 | | app/components/nexus/nexus-tabs.tsx |
 676 | NexusTabButton, NexusTabRail, NexusTabStack | - | actions, tabs, layout |
 NexusTabFrame 3, NexusTabLabel 2, NexusTabRail 2, NexusTabStack 2 | |
 app/components/nexus/nexus-ui.tsx | 1261 | NexusThemedBevelEdges, NexusBevelEdges,
 NexusCard, NexusSectionHeader, NexusBadge, NexusChevronIcon, NexusActionButton,
 NexusSegmentedPill, NexusAttachedTabRail, NexusInlineSelect | - | actions, cards, tabs,
 overlays, forms, layout | NexusActionButton 1, NexusCard 1, NexusSectionHeader 1,
 NexusInlineSelect 1, NexusSegmentedPill 1, NexusTabFrame 3, NexusTabLabel 2, NexusBadge 1 |

Feature component inventory

These files are feature-specific compositions. Most should keep their domain/controller
 logic but gradually consume shared ui/* templates for chrome, forms, overlays, panels, and
 feedback.

| File | Lines | Component candidates | Families | Raw primitives | Shared primitives referenced || --- | ---: | --- | --- | --- | --- ||

app/components/nexus/features/discussions/ | post-inventory extraction | DiscussionFeedPanel, DiscussionThreadPanel, DiscussionPostPanel, DiscussionThreadToolBar, DiscussionFeedPostCard, DiscussionRootPostCard, DiscussionReplyTree, DiscussionVotePill, DiscussionReplyCountPill, DiscussionPostComposer, DiscussionReplyComposer | cards, actions, forms, feedback, layout | Pressable retained only inside feature controls | NexusActionButton, NexusCard, NexusBadge, NexusLoadingBoundary, NexusTextInput, NexusTextArea || app/components/nexus/discussions/nexus-discussion-focus-panel.tsx | 127 | NexusDiscussionFocusPanel | cards, layout | Pressable 3 | NexusActionButton 2, NexusCard 3, NexusBadge 3 || app/components/nexus/focus/nexus-focused-packet-section.tsx | 105 | NexusFocusedPacketSection | cards, layout | Pressable 3 | NexusFocusedPacketSection 1 ||

app/components/nexus/features/sidebar/ | post-inventory extraction | NexusRailToggle, NexusGuestAvatar, NexusPreferenceSwitch, NexusCurrentContextCard, NexusFunctionMenuContent, NexusScopeActionMenu, NexusScopeListRow, NexusGroupedScopeRows, NexusScopeMenuContent | actions, cards, forms, layout, feedback | Pressable and ScrollView retained inside feature controls | NexusCard, NexusCardMenuButton, NexusChevronIcon, NexusLoadingBoundary, NexusThemedBevelEdges || app/components/nexus/features/locality/ | post-inventory extraction | LocalityCreateSearchPanel, LocalityCreateBuilderPanel, LocalityParentPickerDialog, LocalityKindPickerDialog, LocalityCreateKindDialog, LocalitySelectedResultDialog, LocalityRemoveConfirmDialog, LocalityOutcomeDialogs, LocalityCreateGraphRow, LocalityCreatePreviewPanel | overlays, forms, cards, layout, feedback | Pressable/TextInput/ScrollView retained inside feature controls | NexusActionButton, NexusBadge, NexusCard, NexusLoadingBoundary, NexusModalShell, NexusSearchField, NexusSearchResultList || app/components/nexus/nexus-auth-gate.tsx | 566 | NexusAuthGateModal | overlays | Pressable 4 | NexusActionButton 3, NexusCard 3 ||

app/components/nexus/nexus-identity-ui.tsx | 377 | IdentityPageShell, IdentityField, IdentityInput, IdentityRouteLinks, IdentityPreferenceCard, LocationLookupField | cards, forms, layout | Pressable 3, TextInput 3, ScrollView 3 | NexusActionButton 6, NexusCard 7, NexusSectionHeader 2, NexusSegmentedPill 3 ||

app/components/nexus/nexus-packet-explorer.tsx | 772 | NexusPacketExplorer | overlays | Pressable 9, Modal 3 | NexusActionButton 2, NexusCard 3 ||

app/components/nexus/nexus-shell-entry-gate.tsx | 113 | NexusShellEntryGate | overlays, layout | Pressable 2 | NexusActionButton 2, NexusCard 3, NexusBadge 2 ||

app/components/nexus/nexus-shell.tsx | 243 | NexusShell | layout | Pressable 2 | - ||

app/components/nexus/nexus-sidebar.tsx | 881 after feature extraction | NexusSidebar controller; rail/menu/scope/preference components moved to features/sidebar/ | actions, cards, forms, layout, feedback | route still owns profile/prefs drawer shell Pressables and rail ScrollViews | NexusCard, NexusSegmentedPill, NexusThemedBevelEdges ||

app/components/nexus/packet-explorer/nexus-packet-explorer-content.tsx | 768 | NexusPacketExplorerSeededSummary, NexusPacketExplorerLineagePanel, NexusPacketExplorerActionsPanel, NexusPacketExplorerValidationPanel, NexusPacketExplorerContent | actions, layout | ScrollView 3 | NexusActionButton 6, NexusCard 29, NexusBadge 11 ||

app/components/nexus/packet-explorer/nexus-packet-explorer-data-panel.tsx | 210 | NexusPacketExplorerDataPanel | layout | - | NexusCard 11, NexusBadge 7 ||

app/components/nexus/packet-explorer/nexus-packet-explorer-export-panel.tsx | 795 | ExportPreviewCard, NexusPacketExplorerExportPanel | cards, forms, layout | Pressable 3, TextInput 6 | NexusActionButton 6, NexusCard 19, NexusInlineSelect 2, NexusSegmentedPill 2,

Route-local component inventory

Route-local components are the highest-risk extraction zone because they often blend controller state, data fetching, and UI chrome. They should be audited for behavior before any extraction.

| Route file | Lines | Local component candidates | Families | Raw primitives | Shared primitives referenced | | --- | ---: | --- | --- | --- | | src/app/nexus/_layout.tsx | 71 | NexusLayoutContent, NexusLayout | layout | - | - | | src/app/nexus/account.tsx | 203 | NexusAccountPage | forms | ScrollView 3, Switch 1 | NexusActionButton 7, NexusCard 9, NexusSectionHeader 2, NexusBadge 9 | | src/app/nexus/dashboard.tsx | 655 | NexusDashboardPage | overlays | Pressable 2, Modal 3, ScrollView 3 | NexusActionButton 3, NexusCard 11, NexusSectionHeader 2, NexusFocusedPacketSection 2, NexusPreviewPanel 11, NexusActionList 5, NexusActionListItem 3, NexusBadge 5 | | src/app/nexus/discussions.tsx | 1691 after panel extraction | NexusDiscussionsPage route/controller; workspace panels and post/reply/vote feature components moved to features/discussions/* | forms, cards, feedback, layout | route still owns outer page ScrollView composition | NexusActionButton, NexusCard, NexusSectionHeader, NexusTabStack, NexusBadge | | src/app/nexus/identity/claim.tsx | 425 | NexusIdentityClaimPage | support | - | NexusActionButton 8, NexusCard 9, NexusBadge 3 | | src/app/nexus/identity/create.tsx | 328 | NexusIdentityCreatePage | support | - | NexusActionButton 5, NexusCard 3 | | src/app/nexus/identity/restore.tsx | 152 | NexusIdentityRestorePage | support | - | NexusActionButton 5, NexusCard 3 | | src/app/nexus/identity/security.tsx | 588 | NexusIdentitySecurityPage | support | - | NexusActionButton 12, NexusCard 25, NexusSegmentedPill 3, NexusBadge 11 | | src/app/nexus/identity/sign-in.tsx | 639 | NexusIdentitySignInPage | support | Pressable 3 | NexusActionButton 7, NexusCard 13, NexusTabRail 2 | | src/app/nexus/index.tsx | 14 | NexusIndexPage | support | - | - | | src/app/nexus/library.tsx | 317 | NexusLibraryPage | support | ScrollView 4 | NexusActionButton 5, NexusCard 9, NexusSectionHeader 2, NexusBadge 5 | | src/app/nexus/locality/create.tsx | 2470 after feature extraction cleanup | NexusLocalityCreatePage route/controller; search/build panels, picker dialogs, graph rows, preview panels, and obsolete level-row UI moved or removed | overlays, forms | route owns graph state, loading, mutations, auth, and outer scroll composition | NexusCard, NexusSectionHeader, NexusTabRail, NexusBadge | | src/app/nexus/roles.tsx | 714 | NexusRolesPage | forms | TextInput 2, ScrollView 3 | NexusActionButton 7, NexusCard 23, NexusSectionHeader 2, NexusTabRail 2, NexusBadge 17 | | src/app/nexus/trust.tsx | 659 | NexusTrustPage | forms | TextInput 2, ScrollView 3 | NexusActionButton 9, NexusCard 31, NexusSectionHeader 2, NexusBadge 20 | | src/app/nexus/votes.tsx | 210 | NexusVotesPage | support | ScrollView 3 | NexusActionButton 4, NexusCard 13, NexusSectionHeader 2, NexusBadge 7 |

Support and context files with UI adjacency

These are not primary UI templates, but several contain providers, controllers, or state that UI templates depend on. They should remain outside presentation folders unless their visual pieces are separated first.

| File | Lines | Component/provider candidates | Exported types | Zone | | --- | ---: | ---
| --- | --- | | app/components/nexus/identity-shell-context.tsx | 2056 |
IdentityShellContext, IdentityShellProvider | - | support | |
app/components/nexus/identity-shell-fortress-adapter.ts | 367 | - | - | support | |
app/components/nexus/nexus-auth-gate-types.ts | 12 | - | - | support | |
app/components/nexus/nexus-feature-status-context.tsx | 234 | NexusFeatureStatusContext,
NexusFeatureStatusOverlay, NexusFeatureStatusProvider | - | support | |
app/components/nexus/nexus-feature-status-registry.ts | 203 | - | NexusFeatureStatusKind,
NexusFeatureStatusEntry, NexusFeatureStatusId | support | |
app/components/nexus/nexus-route-utils.ts | 77 | - | - | support | |
app/components/nexus/nexus-shell-chrome-context.tsx | 56 | NexusShellChromeContext,
NexusShellChromeProvider | - | support | | app/components/nexus/nexus-shell-context.tsx |
859 | NexusShellContext, NexusShellProvider | - | support | |
app/components/nexus/nexus-shell-preferences.ts | 67 | - | NexusShellPreferenceMode,
PersistNexusElementPreferenceInput, PersistNexusElementPreferenceResult | support | |
app/components/nexus/packet-actions/nexus-packet-action-registry.ts | 57 | - |
NexusPacketActionInput, NexusPacketActionHandlers, NexusPacketActionRegistryInput | support
|

Extraction candidate queue

Recommended order, based on duplication, risk, and likely payoff:

1. Overlays and modal shells: initial shared modal shell and outcome/confirm primitives now exist under ui/overlays; remaining work should focus on auth/session gates, feature-status overlays, badge tooltips, and any modal content that deserves picker/outcome-specific composition. Keep auth/session logic separate from modal chrome.
2. Form field shells and searchable result lists: base segmented, inline-select, field shell, text input, text area, action row, and search-result primitives now live under ui/forms; Explorer search/export/import fields, identity lookup, locality create search dropdowns, trust notes, roles comments, and discussion composers now have shared presentation seams. Remaining work should target preference rows and broader route-local form sections.
3. Layout frames and section scaffolds: page/scroll frames, panel/workbench wrappers, panel headers, section bands, toolbar rows, and metric grids now live under ui/layout; identity page shell, dashboard/trust/roles metric rows, and a Packet Explorer workbench panel have begun adopting them. Remaining work should target repeated route page wrappers, packet explorer panel shells, sidebar rail sections, and dashboard card rows.

4. Feedback states: scoped loading and basic empty/error/warning/status/operation cards now live under ui/feedback; remaining work should replace route-local loading, refreshing, empty, and outcome copy only where the shared card can preserve the existing layout.
5. Tab unification: keep Explorer document tabs as an extension of shared tab primitives. Do this after overlays/forms/layout so tab work does not become a broad behavioral rewrite.
6. Packet inspection/focus panels: focus, preview, dashboard, library, and Explorer all present packet summaries with related action/badge/provenance slots. Extract only after action/menu behavior stays stable.
7. Discussion feature promotion candidates: workspace panel shells, scrollable feed panels, thread toolbars, composers, vote/reaction pills, and recursive tree rails are now isolated under features/discussions; promote them into ui/* only after another surface needs the same skeleton.
8. Sidebar feature promotion candidates: rail toggles, compact nav rows, preference switch rows, grouped collapsible sections, scope/action row shells, and anchored compact action menus are now isolated under features/sidebar; promote them into ui/* only after another surface needs the same skeleton.

Suggested future folder map

“txt app/components/nexus/ui/ actions/ # buttons, menu buttons, action menus, action lists, badge strips cards/ # base surfaces, packet cards, focus/preview cards, stat cards, warning/outcome cards tabs/ # shared tab frame/label/rail/stack plus Explorer document-tab extensions overlays/ # modal shell, confirmation, outcome dialog, popover, gates, tooltip hosts forms/ # field shell, text input, text area, composer, search box/result list, picker shell, toggles layout/ # page/scroll frame, section band/header, panel/workbench panel, panel split, rail section, toolbar frame, metric grid feedback/ # loading, empty, error, warning, operation result/status rows “

Guardrails for later migrations

- Preserve visuals and behavior first; extract structure before changing design.
- Do not move runtime/core logic into shared UI folders.
- Do not make packet-type-specific UI decisions inside generic templates. Generic templates receive slots, labels, state, handlers, and scope strings.
- Prefer one migration family per pass. A pass that touches overlays, tabs, forms, and route controllers at once is too wide.
- Update this hard inventory or the component catalog after each consolidation pass so the map does not rot into wall art.

Core Entities And Packet Model

Packet graph foundation

The packet graph remains the most robust unifying model.

Core packet rules:

- packet_id is the stable logical packet identity
- revision_id is the immutable exact revision identity
- parentrevisionrefs represent DAG ancestry
- schema_version tracks type-shape compatibility
- edges are the shared typed relationship collection
- authorityscoperef and applicablescopecrefs stay separate
- revision_mode expresses type write semantics

Raw stored packets remain the historical signed fact. Adapted packets are runtime read views, and interpreted or read-model outputs are additional projections on top of those reads.

Core entities

Element

The universal identity anchor.

- person
- assembly
- organization
- service
- overlay or coordination group

Current direction:

- Element remains the durable actor, scope, and container type
- Element.subtype is the canonical classifier surface
- fresh Element packets reject the old top-level kind classifier
- dotted subtype forms such as person.claimed_identity or assembly.city are the forward model when the older shape previously depended on multiple classifier fields

Scope

The context or lens around an element. Every person can be treated as their own scope lens through You.

Assembly

A policy-governed civic or geographic element subtype, not a separate storage primitive.

Initiative

A generic Nexus concept for policy, template lineage, defaults, and work hierarchy. The pre-reseed reset treats Action(subtype: initiative) as the canonical top-level initiative anchor above campaign, program, mission, and task semantics. Cause is no longer an active fresh packet type.

Claim

The assertion and argument type.

Current forward direction:

- Claim can target packets generically
- Claim may carry claim_markdown and supporting refs
- Claim(subtype: relation_assertion) is the forward shape for claims specifically asserting a relation
- relation-specific claim semantics live under Claim(subtype: relationassertion).relationassertion.subtype

Current home-locality direction:

- canonical writes now use relation.residence.add
- that write produces Relation(subtype: residence) only; claims and reactions can be added around the relation, but are not automatically minted
- legacy residence.claim.set is retired from live fresh writes; historical Claim(residence) material remains readable through compatibility projections
- revise and withdraw semantics remain packet-native: status changes are represented by newly signed packet material rather than in-place mutation

Reaction

The evidence, certification, support, dispute, and packet-signal type.

Current code truth:

- claim support and dispute currently stay on Reaction
- Reaction and Claim are still distinct in code
- Reaction.subtype is the canonical reaction classifier

Role

A reusable role definition packet type. Exact-scope role assertions are currently represented through Claim(subtype: "relationassertion") with relationassertion.subtype: "role_association".

Policy

The configuration and legitimacy type for defaults, thresholds, and later execution rules.

Current direction:

- actual adopted or followed graph facts belong in Relation
- asserted or disputable statements belong in Claim
- evidence and support/dispute posture belong in Reaction
- legitimacy-sensitive relation rules belong in Policy.relation_requirements
- default inheritance belongs in Policy.default_policy, using packet refs for policies, templates, default packet sets, and preference material rather than runtime-only default names
- governance readiness belongs in Policy.governance_policy, reserving voter eligibility, trust stage, quorum, approval, vote method, and decision-report hooks without executing voting yet
- dependency requirements remain in Policy.dependencies_policy; subscriptions record what a subject accepts or excludes, and projections compare the two

Decision

A governance artifact type that exists in infrastructure and read surfaces, but whose real workflow semantics are still developing.

Converging civic grammar

The long-term Nexus direction is converging on a smaller reusable civic grammar:

- Element
- Relation
- Claim
- Reaction
- Report
- Action
- Policy

Current code already implements most of that grammar directly. Action now carries the forward initiative/work hierarchy and packet-backed default override refs, while Report has landed as a real packet type but its live use is still intentionally narrow.

Current code now ships the first concrete Report type use:

- Report(subtype: verification_report)
- Report(subtype: import_report)
- Report(subtype: decision_report) as reserved governance closure-report shape

This first live usage should still be understood narrowly. The type now exists as real packet infrastructure, but broader audit, incident, decision, or resolution report semantics remain later architectural work rather than implicit current product truth.

Packet-backed defaults

Defaults should remain packet-native rather than becoming Element fields or runtime constants. The current pre-reseed inheritance direction is:

- Definition(subtype: defaults_definition) parts carried alongside each packet definition
- bundle/default packet-set material
- initiative Action policy/template/default packet-set refs
- element policy/template refs or relations
- actor/client Preference

OWA-specific behavior should be represented by the default OWA Action(subtype: initiative) packet and its linked policies, templates, bundles, and preferences, not by special cases in generic packet schemas.

The default OWA seed now links its forward Action(subtype: initiative) anchor to default-inheritance and governance-baseline Policy packets. New default/policy resolution should use the Action anchor, then layer policy-selected defaultsdefinitionrefs and explicit overrides on top of definition-native defaults.

Schema evolution discipline

Before changing packet schemas, read this chapter first.

Also read docs/implementation-guide/trust-moderation-and-policy.md before changing Claim, Reaction, Relation, or Policy semantics.

For any packet type schema version change, the required checklist is:

- update the active schema or body shape
- update the type compatibility registry entry
- add or update upcast and downcast adapters where backward compatibility is intended

- update current schema version metadata
- update builders and type build definitions so new writes emit the canonical current shape
- update signature and write-preparation behavior if additive or defaulted fields affect compatibility or signing
- add or update tests for parse and read compatibility, adapted read behavior, write preparation, and any supported upcast or downcast path
- document the change in the relevant chapter and add a decision-log note when the change is architecture-significant

Relationships and graph semantics

The graph should continue to express relationships through typed refs and edges rather than UI-only linkage. Important relationship categories remain:

- references or depends_on
- proposes or supersedes
- belongs_to
- scoped_to
- endorsedby or vouchedby
- implements or enacts
- reports_on
- forkof or derivedfrom

Current scope-graph direction:

- canonical mounted ancestry prefers Relation(subtype: defaultancestryparent)
- canonical home-locality projection prefers Relation(subtype: residence); legitimacy evidence can attach through separate Claims and Reactions when policy or contestation requires it
- canonical follow relations now use Relation(subtype: follow) and are actor-only; legacy shell follow preferences are compatibility-only read input
- canonical association now uses Relation(subtype: association) without automatic claim wrapping, and associated scopes now count as mounted related scopes in shell projection
- canonical policy adoption now uses Relation(subtype: subscription) targeting a Policy packet rather than a separate adopts_policy relation subtype
- dependency requirements remain policy-layer semantics, usually Policy.dependenciespolicy, while dependson remains available only as a structural edge type rather than a Relation subtype

- subscription relations can carry subscription_options for inherited/default policies, dependencies, modules, templates, and default packet sets; excluding a required default does not erase the subscription, but projection should report partial alignment or review needs
- Relation(subtype: definedbylocation) is the live read seam for linked Location packets
- locality.path.create now emits locality Element packets, defaultancestryparent relations, provisional Location(subtype: region) packets, and definedbylocation relations together
- locality rows can now carry dynamic descriptor metadata, which is currently stored in linked Location.spatialpayload.scopeddescriptor rather than through a packet schema bump
- legacy locality levels such as nation | region | city | district now function as compatibility buckets, while actual ancestry comes from the ordered path graph and not from a hardcoded four-slot ladder
- locality depth remains projection-only and should not be stored as a universal packet truth
- legacy parent_scope ancestry, shell follow preferences, and legacy Claim(residence) reads now belong in explicit compatibility projections or compatibility mirrors rather than inline main-path logic

Reaction

Reaction is the current packet type for target-agnostic responses. It replaces the prior split between standalone vote packets and reaction packets.

A reaction can carry independent channels for signal aggregation, support/dispute posture, and emotional response:

- vote_value: basic up/down signal used for visibility and formal vote-like aggregation
- attestation_value: support or dispute only
- emoji_keys: small emoji-key set for how people feel about a target

Relations, roles, proposals, discussion packets, and future action packets can all be reaction targets without the reaction needing packet-type-specific semantics.

Trust, Moderation, And Policy

Trust direction

Trust should remain layered, inspectable, and threshold-based rather than collapsing into a hidden score.

Current code truth:

- trust stages are currently explicit and threshold-based

- support and dispute evidence are packet-backed
- role posture is projected from scoped claims plus claim-targeted reactions

Direction:

- trust should stay scope-local
- trust should gate posting, voting, review, moderation, or role authority only through explicit policy and thresholds
- evidence and posture should remain inspectable even when defaults hide or deprioritize low-trust activity

Moderation and automoderation

Status: canon candidate

Current code truth

- moderation workflows are not yet fully implemented
- some read surfaces already expose packet-backed evidence and action visibility
- disabled placeholders are visible in several surfaces

Direction

- automoderation should be a generic feature, not a route-local trick
- default moderation baselines should come from the relevant element's policy
- user-level moderation preferences should be available as adjustable policy or settings choices, similar in spirit to write protection preferences
- moderation should primarily act as visibility projection over packet and reaction graphs before it becomes hard rejection logic

Unresolved

- which moderation controls belong in generic policy versus OWA default policy
- which interactions should remain soft projection versus hard runtime enforcement
- how much moderation preference state should be actor-level policy versus client-level shell preference

Reactions, attestations, and claims

Status: canon candidate

Current code truth

- Relation is the structural type for adopted graph facts

- Claim is its own live packet type for assertions, arguments, and disputable relation claims
- Reaction is its own live packet type for evidence, certification, support/dispute, and packet-signal responses
- current discussion voting uses packet-signal reactions rather than a separate reaction type
- current claim support and dispute still run through the reaction service and reaction indexes

Direction

- keep Claim and Reaction distinct
- treat Claim as the forward layer for relation assertions and other packet-targeted arguments
- treat Reaction as the forward layer for supporting, disputing, certifying, or signaling around packets, including Claims and Relations
- do not require Claim wrappers for every Relation
- let policy require supporting Claims for legitimacy-sensitive Relations where needed

Current implementation note:

- the new `Policy.relation_requirements` seam exists so stricter relation-support rules can be expressed generically instead of being hardcoded as route-only logic
- `Policy.defaultpolicy` now carries packet-backed default refs for policies, templates, `Definition(subtype: defaultsdefinition)` packets, default packet sets, and preference material, plus explicit override paths; it must not introduce runtime-only default labels
- `Policy.governance_policy` now reserves packet-backed governance hooks for voter eligibility, minimum trust stage, quorum, approval threshold, vote method, and decision-report expectations
- this chapter should be read before changing Claim, Reaction, Relation, or Policy semantics because it owns the intended separation between assertion, evidence, graph structure, and policy requirements
- packet-native follow does not currently require a supporting claim in this phase
- packet-native association no longer auto-mints a supporting self-issued `Claim(subtype: relation_assertion)` alongside the structural relation

Current home-locality policy note:

- OWA-sensitive home-locality legitimacy resolves through `Action(subtype: initiative)` as the forward OWA policy/default anchor
- a canonical `Relation(subtype: residence)` is enough for default mounted home-locality projection; stricter `Policy.relation_requirements` can still require extra evidence for specific scopes

- the expected support model in this phase is separate Claims and Reactions attached around Relations only when evidence, dispute, or policy asks for them
- legacy claim-only home-locality reads remain compatibility projections, not the forward legitimacy model

Current dependency authority note:

- Definition dependencies_definition parts describe packet requirements and local engine contracts; runtime registries only validate and interpret those refs
- workflow dependency IDs must resolve to a Definition dependency part, Policy semantic, operation ontology entry, workflow resolver allowlist, or trusted local engine contract
- trusted runtime capability metadata is allowed only when it points back to packet meaning or an explicit local engine contract

Regulation and policy resolver direction

Status: initial trusted coordinator foundation exists; full governance language remains future work.

Policy packets exist today, and the Trusted Regulation Coordinator now provides the gated runtime surface for policy-oriented resolution. It resolves regulation contexts, policy contexts, write-policy gates, requirement lists, and readiness audits. Defaults and dependencies are now owned by the Trusted Planning Coordinator because they shape construction plans rather than enforce policy by themselves. Full proposal, voting, moderation, and role-gate semantics still need richer policy descriptors before governance becomes interactive.

The regulation resolver is designed to answer, progressively:

- who can create or mutate each packet type in a scope
- who can propose, vote, attest, review, or moderate
- which trust stage, role, or association gates apply
- which quorum, threshold, and vote method applies
- which policy-backed defaults are allowed or required when Planning asks
- whether proposal discussions, decisions, or decision reports should be created
- which packet types are allowed in a given scope

This resolver stays separate from builders and planners. Builders shape packets. Planning applies defaults, checks structural dependencies, selects builders, and composes operation plans. Regulation classifies the active policy envelope around an operation: allowed, required, blocking, advisory, definition-audit-only, or future-hook. Policy resolution is intentionally usable outside packet creation, including projection, import/export review, moderation, runtime reads, and governance checks.

Longer-term civic review framing

Direction:

- Claim should be understood as an unresolved assertion or request that invites evaluation, scrutiny, or action
- Reaction should be understood as a signed stance toward a target, such as support, dispute, certification, rejection, or abstention
- formal voting should route through governed Reaction aggregation inside a recognized process with eligibility and policy rules
- Report should be the long-term home for findings, outcomes, and context around a target
- a resolution or decision artifact should eventually read as a process-backed closure report rather than as unexplained authority
- quorum, minimum trust, eligibility, approval thresholds, and voting gates should be policy-backed defaults that can be inherited from the applicable initiative Action or scope policy and overridden by explicit packet refs

This framing is architectural direction, not a statement that all of those packet types or workflow surfaces already exist as live runtime behavior.

Verification truthfulness direction

Future verification UI and policy language should distinguish at least:

- structurally valid
- cryptographically verified
- provenance known
- locally trusted
- substantively true

Current product language does not yet fully expose those distinctions. The next verification/reporting work should avoid collapsing them all into one vague verified concept.

Current verification chapter truth:

- the runtime now signs verificationreport and importreport packets with a normal local validator identity
- those report packets are the signed record of what the node concluded
- a local runtime-owned verification index/cache exists so dashboard cards, Explorer search rows, export lookup rows, action menus, and Explorer verification views can project truthful current status quickly

- imported external verification reports are readable evidence, but local trusted status still comes only from the latest local validator report in this phase

Governance, Initiatives, And Decisions

Initiatives

Status: canon candidate

Current code truth

- initiative-specific filtering, subscriptions, and version visibility are not yet active product behavior
- current code does not yet implement initiative-version subscriptions or official versus unofficial filtering

Direction

- Action(subtype: initiative) is the forward top-level initiative packet for policy, template lineage, default packet sets, and later work hierarchy
- OWA should be modeled as an initiative Action inside Nexus, not as a hardcoded exception
- Cause is no longer an active fresh packet type
- lower work hierarchy levels should be represented through Action subtypes such as campaign, program, mission, and provisional task
- "official OWA" should mean conforming to recognized OWA dependencies, templates, and policies
- dependency requirements remain policy-layer semantics; depends_on is not a Relation subtype
- policy adoption is modeled by Relation(subtype: subscription) targeting a Policy, not by a separate adopts_policy relation subtype
- assemblies remain valid Nexus objects even when they fork or diverge from canonical OWA lineage

Unresolved

- the exact packet/API shape for initiative conformance and recognition beyond Action refs and policies
- how official, unofficial, derived, and forked initiative states should be encoded

Initiative versioning and subscriptions

Status: canon candidate

Direction

Initiative lineage should support version-aware adoption and viewing, including:

- current official version
- older official versions
- upgrade availability
- optional inclusion of unofficial forks

Subscriptions and default visibility should eventually support choices like:

- current official only
- all official versions
- official plus unofficial
- multiple initiatives together
- inherited default policies/dependencies with explicit deselection and visible partial-alignment states

Unresolved

- the exact version packet or metadata shape
- whether subscriptions belong primarily to actor policy, shell preference, or both
- how version compatibility and upgrade prompts should surface in Explorer and feeds

Assemblies, custody, and legitimacy

Status: canon candidate

Direction

- assemblies are elements with policy-governed civic semantics, not owned subsidiaries
- custody, legitimacy, policy authority, and governance outcome should remain distinct concepts
- association should remain a simple claim first; typed categories are unsupported
- multi-key assembly authority, custody transition, and protected-state mutation should remain explicit later design work

Unresolved

- the exact custody and multi-key packet model
- how authority grants or keyset policies should be represented
- how passed decisions eventually authorize protected state changes

Governance resolver direction

Status: canon candidate.

Governance should be built on packet-backed policy resolution, not hardcoded proposal or vote fields. Assembly-capable elements can receive default governance policies through Nexus/OWA default profiles. Those policies enable proposal behavior, define voting rules, and determine whether proposal discussions, decisions, or reports are generated by default.

The intended chain is:

- an assembly-capable element receives or adopts a governance policy
- the governance policy enables proposal creation and proposal discussion defaults
- proposal creation resolves the active governance policy before packet candidates are built
- votes or vote-like reactions target the proposal under that policy
- a decision packet records the formal outcome
- a decision report may record tally, evidence, and process context when policy requires it

OWA can provide governance presets and default policies. OWA should not override Nexus builders or redefine proposal, reaction, decision, or report packet anatomy.

Proposals and decisions

Status: canon candidate

Current code truth

- votes remain read-only
- proposals, votes, and decisions are not yet a fully interactive trusted workflow

Direction

- decisions should begin as legitimacy records and civic signals
- policy may later allow some decisions to activate real effects
- proposals should declare their intended outcome as explicitly and programmatically as possible
- quorum, eligibility, minimum trust, approval thresholds, and voting gates should come from packet-backed Policy defaults attached through the applicable initiative Action, scope, or proposal context rather than from hardcoded Vote fields
- a Decision should remain the formal outcome artifact, while Report(subtype: decision_report) is reserved for evidence, tally, and process-context closure material

Proposal outcome categories should eventually include:

- policy adoption

- mission or coordination plan
- resolution
- signal or poll
- other explicit action packages

Automatic execution should only be described as future behavior for explicitly supported built-in action handlers or canonical approval artifacts.

Unresolved

- which decision types can become effect-bearing first
- how multi-key assembly authority interacts with execution
- whether execution produces direct mutations, canonical approval packets, or both

Decision Log 2026-04

This monthly log condenses the April 2026 decisions that remain most important for current implementation and future migration work.

2026-04-08 foundations

- Packet identity uses stable packetid plus immutable revisionid.
- Revision history is a DAG with parentrevisionrefs, not a single chain.
- Packet relationships normalize into one typed edge collection.
- Element remains the identity-root type for people, assemblies, organizations, and services.
- The dedicated Nexus shell lives under /nexus/*.
- Function-first and scope-first remain one system rather than two route trees.

2026-04-09 shell and discussion shape

- Shared browser and Nexus query services over the packet graph became the projection seam.
- Active shell and scope routes moved onto packet-backed payloads instead of mock arrays.
- Discussions became a connected forum shell with feed, thread, and post workspaces.
- Feed cards became the primary thread-launch surface.
- Cursor paging and reply-branch controls became first-class route behavior.

2026-04-10 identity and auth

- Every graph-writing actor became a cryptographic Element(kind: "person"), including guests.

- Claimed auth uses local keys plus server challenge sessions rather than server-owned identity truth.
- Claimed auth now supports passkeys, rotating sessions, and explicit write-approval modes.
- Identity ceremonies moved into the Nexus shell.
- Reaction replaced packet votes as the reusable trust edge.

2026-04-17 architecture and trust pivot

- The final source roots became core, runtime, and app, with src/app as the Expo Router shell.
- Trust replaced Account as the top-level function surface.
- You became a first-class personal scope lens.
- Roles became a dedicated scope workspace.
- Role, Claim, and packet compatibility infrastructure became first-class runtime concerns.

2026-04-18 to 2026-04-20 locality and identity hardening

- Legacy signed identities remained valid after the roles-and-claims refactor.
- Home locality became the mounted geographic truth, while follow became shell preferences.
- Locality search and locality creation hardened around canonical geographic assembly flows.
- Location disclosure remained separate from mounted home-locality truth.

2026-04-23 and 2026-04-24 compatibility and fortress corridor

- Schema compatibility moved into a real raw/adapted/read-for-write pipeline.
- Fortress pass 1 moved discussion write admission into core mutation law.
- Fortress pass 2 introduced the shared prepare/sign/finalize corridor and packet-backed write-lock policy updates.

2026-04-25 to 2026-04-29 packet floor and verification

- Packet compatibility coverage is now classified explicitly.
- Legacy write seams were sealed before discussion-type cleanup.
- Raw packet signatures verify before adaptation.
- Live and near-live packet types now share one generic builder floor.

2026-04-30 surface-level runtime contracts

- Discussion workspaces now project action affordances through the shared NexusActionState and NexusActionIntentDescriptor contract.

- Packet Explorer became a shell-level read-only inspection workspace.

Decision Log 2026-05

2026-05-19 initiative action and discussion schema readiness

- Pre-reseed initiative semantics now point to Action(subtype: initiative) as the forward top-level initiative/work hierarchy above campaign, program, mission, and provisional task semantics.
- Action(subtype: initiative) is the fresh-reseed initiative/default anchor; Cause is pruned from active canon.
- Canonical discussions now reserve Discussion(subtype: post) for top-level forum artifacts that start threads, while reply chains use Discussion(subtype: message).
- Governance schema hooks remain policy-backed: quorum, minimum trust, voting gates, and decision-report closure semantics should be expressed through linked Policy/default material, Decision, and Report(subtype: decision_report).

This monthly log condenses the May 2026 decisions that remain most important for current public-surface and documentation shape.

2026-05-01 public surface convergence

- Shared public actions and section shells now route through one surface contract.
- PublicCardFrame separates shared graphic treatment from page-owned content layout.
- Support and Docs now share the same public frame and panel system.
- Public card frames gained ambient generated backgrounds owned by the shared frame layer.
- About surfaces joined the shared public frame path without absorbing card-specific animation rules.
- Public sections now use the universal card animation path.
- Public theme cleanup centralized lingering route-local styling values into named public seams.

2026-05 chaptering follow-up

- Canon docs are being split into index files plus shallow chapter folders to reduce edit cost, preserve top-level link stability, and keep current truth, durable architecture, and roadmap work separated cleanly.

2026-05 docs workflow cleanup

- The multi-chapter internal docs now treat chapter files as the canonical content source, while the top-level files remain short local index shells only.
- Public docs generation for implementation guide, specifications, and roadmap now compiles from chapter files only instead of concatenating the top-level shell files into the public artifact.
- The docs build now validates chaptered manifest entries, missing or duplicate source files, generated-artifact misuse, and shell-file drift before producing readable or downloadable outputs.

2026-05 shell honesty foundation

- Nexus now uses a shell-level early-access gate as a session-scoped entry warning rather than leaving public-facing instability as unstated product lore.
- The gate is intentionally lightweight and UI-owned: it is not packet-backed, not actor-specific, and not yet a tutorial flow.
- The same shell seam should later be able to host onboarding, release notes, or other blocking notices without redesigning the overlay stack.

2026-05 scoped Nexus loading foundation

- Nexus loading now has a UI-owned provider, boundary, overlay, and hook for caller-selected visual scopes.
- The loading layer intentionally ignores packet type, action registry identity, mutation intent, and runtime operation categories; UI boundaries decide what visual area is busy.
- A scope becomes active immediately to block duplicate input, while the visual overlay is delayed and minimum-duration guarded to avoid spinner flicker.
- Shared buttons and action-menu items can now opt into scoped loading with caller-supplied scope strings, while packet action registries and runtime operations remain loading-agnostic.

2026-05 Nexus UI folder foundation

- Nexus UI consolidation began with safe shared-module moves only: loading moved under `ui/feedback/loading`, action cards under `ui/cards/action-card`, action lists under `ui/actions/action-list`, and shared tab primitives under `ui/tabs`.
- The move establishes the canonical `app/components/nexus/ui/*` home without splitting `nexus-ui.tsx`, changing visuals, or moving feature-specific Explorer, locality, discussion, dashboard, preview, or focus components.

2026-05 Nexus overlay consolidation start

- Nexus overlay consolidation began with reusable modal chrome under `app/components/nexus/ui/overlays/*`, including a modal shell plus outcome, confirmation, and popover primitives.
- Dashboard packet validation, packet Explorer import outcomes, and locality create/reuse/picker dialogs now share the same modal shell while preserving their existing visual classes, close behavior, and route-local content.
- Auth/session gates and feature-status overlays remain specialized for later migration so generic overlay UI does not absorb session or shell behavior.

2026-05 Nexus search UI consolidation start

- Nexus search presentation now has shared primitives under `app/components/nexus/ui/forms/search/*` for search fields, result lists, result rows, status text, empty/error states, and loading-aware result boundaries.
- The consolidation is UI-only: Packet Explorer, identity, and locality create keep their own debounce timing, API calls, ranking, filtering, selection, and create-candidate behavior.
- Search loading scopes attach to the visual result surface, such as a dropdown or dedicated results panel, instead of packet types, route names, or search engine semantics.

2026-05 Nexus forms and feedback alignment start

- Nexus forms now have shared field shell, text input, text area, and field action-row primitives under `app/components/nexus/ui/forms/*`.
- Nexus feedback now has shared inline loading plus empty, error, warning, status, and operation-card primitives under `app/components/nexus/ui/feedback/*`.
- Identity fields, Packet Explorer import/export fields, trust association notes, and roles support comments began migrating to those primitives while preserving existing route behavior and keeping scoped loading caller-owned.

2026-05 Nexus layout and panel foundation start

- Nexus layout now has shared page/scroll frames, panel/workbench wrappers, panel headers, section bands, toolbar rows, and metric grids under `app/components/nexus/ui/layout/*`.
- Identity page chrome, dashboard/trust/roles metric rows, and a Packet Explorer search workbench panel began adopting these wrappers while preserving current visuals.
- Panel loading remains caller-owned by visual scope; layout components only provide a clean boundary mount point.

2026-05 Nexus UI primitive split

- The broad `app/components/nexus/nexus-ui.tsx` primitive file is now a compatibility bridge over focused `app/components/nexus/ui/*` component-type modules.
- Cards, actions, badges, forms, layout/chrome helpers, and attached tab rails now have canonical family homes while preserving their existing classes and behavior.
- Nexus callers have moved to the canonical `@app/components/nexus/ui` import path; future work should not add new primitive imports from `nexus-ui.tsx`.

2026-05 feature-status explainers

- Disabled and partial Nexus controls now route through a centralized feature-status registry instead of scattering one-off placeholder copy through route files.
- The first-pass status taxonomy is intentionally small: `comingsoon`, `readonly`, `partial`, and `custom`, with registry-owned overrides when a control needs more specific wording.
- Nexus shell now hosts one explainer card overlay at a time so disabled controls can stay compact while still opening truthful contextual explanations above Explorer, Library, Votes, and future surfaces.
- The shared `NexusActionButton` seam remains semantically disabled for these controls, but can now open an explainer instead of acting like a dead button when a feature-status id is attached.

2026-05 Explorer and shell mobile cleanup

- `NexusInlineSelect` now renders its open menu on a true overlay layer instead of trusting local stacking order, so Explorer View as menus can sit above adjacent header controls.
- Packet Explorer tab decks now behave like normal wrapped rows until they exceed three rows, and only then become scrollable.
- Narrow-screen shell entry now prioritizes immediate access to the real menu rails by opening both rails when the tray is opened, instead of preserving collapsed desktop rail state inside the mobile tray.
- That same mobile tray now sizes itself to the currently visible rails and closes entirely when both rails are minimized.
- Library packet highlighting is now treated as Explorer-linked state rather than a permanently sticky URL-only cue, so stale selection clears when the corresponding Explorer packet tab disappears.
- Explorer resize now lives in a dedicated Explorer-specific desktop drag controller rather than the main overlay coordinator, and uses global window mouse tracking plus temporary drag capture instead of a moving `PanResponder` handle.
- Shared browser packet projections now route through the core packet-title projection path instead of falling back directly to raw packet ids, which keeps Explorer and other packet-inspection surfaces from flipping loaded titles into percent-encoded route strings.

- Explorer reading now uses one primary content scroll surface with local collapsible header bands, while Close tabs has moved out of the shell command row and into the packet-tab deck as a packet-management utility.
- Explorer collapse controls were then tightened so the bands do not reserve their own chevron rows: collapsed sections disappear fully, and their restore actions move into the top shell command row as Packets and Views.

2026-05 Packet Explorer export workbench

- Packet Explorer Home is now explicitly split into Search and Export sub-tabs so future Explorer capabilities can grow from one stable workspace without collapsing back into the main packet inspector.
- The first live portability seam is export only: Search stays present but non-live, while Export becomes the active raw-json workbench.
- Packet export defaults to Raw packet, which means the current preferred revision envelope only; bundle wrapping is an explicit opt-in instead of the default artifact shape.
- Bundle export remains additive and transport-oriented: the envelope now records export metadata such as mode, root refs, counts, title, and note without changing packet schemas or mutating the packet envelopes inside packets.
- Phase 1 bundle scopes stay concrete and bounded to current graph/scoping contracts: packet history, outgoing references, incoming referrers, scope-stack ancestry, their combined union, and full local-store export as a separate node-level snapshot tool.
- Preview and download now share one export builder seam with explicit size guardrails so Explorer can show inline JSON only for small payloads while still allowing larger .json bundle downloads without inventing a second export format.

2026-05 Packet Explorer import workbench

- Packet Explorer Home now expands to Search, Import, and Export, with Search-card import shortcuts routing into the new Import workspace rather than spawning separate packet-vs-bundle pages.
- Import is intentionally a two-step flow: Analyze first, then Commit, so pasted or uploaded JSON does not mutate the local store until the runtime preview says the payload is structurally safe.
- Phase 2 import stays generic and backward-tolerant by accepting raw packet envelopes, exported bundle envelopes with packets, legacy bundle envelopes with revisions, and raw arrays, all normalized onto the existing bundle-import substrate.
- Web import now includes a browser .json file picker, but paste remains the universal fallback so the workflow still works outside the browser without adding a cross-platform file abstraction in this pass.

- Import analysis is structural rather than trust-verifying: it reports invalid entries, duplicates, missing parent revisions, type conflicts, affected packets, and likely open targets, then blocks commit on unsafe inputs instead of offering partial-import toggles.
- Post-import repair now preserves local preferred-head intent when divergence appears: if a newly imported branch creates multiple heads and the old preferred head still exists, Explorer restores that preferred revision instead of silently replacing it with the last imported head.

2026-05 Packet Explorer live search and export lookup

- Packet Explorer Search is now a real manual discovery workspace instead of a placeholder card, and its state stays Home-owned so query text, grouped results, and the active category survive normal Explorer tab movement while the session stays open.
- Search intentionally stays preferred/current only in this pass: it reads from the shared packet search index rather than adding FTS, graph traversal search, or historical revision fanout into the main result set.
- Exact revision-id search is still supported through a separate revision resolver seam on the packet store, which lets Explorer map historical revision IDs back to the owning packet while keeping the displayed packet card anchored to the current preferred revision surface.
- Search results are grouped into Direct, Name, and Text, with deterministic ranking and lightweight Open / Export routing instead of turning the first live pass into a broader action hub.
- Export now includes a compact active lookup when no packet target is preloaded, but that lookup intentionally remains narrower than the full Search workspace so Explorer keeps one discovery-oriented search surface and one operational packet-picker.

2026-05 Packet Explorer workspace cleanup

- Home workspace navigation has now moved out of the Home body and into the shared inspector row, so Search, Import, and Export behave like first-class Explorer modes rather than nested content tabs.
- The packet primary rail and the Home workspace rail now share one compact attached-tab treatment, dropping the previous secondary tab copy so the inspector band can stay horizontally accessible without running off-screen.
- Home-only dead controls were removed instead of being left as placeholder chrome, and the Search workspace now keeps only its core manual finder actions rather than duplicating Import navigation through extra shortcut buttons.
- Packet export was re-centered around direct packet lookup when no target is preloaded, with the packet-picker promoted ahead of explanatory copy so the export workspace feels operational instead of instructional.

- Search now defensively tolerates a missing runtime revision resolver by falling back to the preferred/current packet index path instead of failing the whole search request.

2026-05 Packet Explorer final polish

- Link traversal in Explorer is now explicit instead of ambiguous: grouped links expose Open in new tab, Open in current tab, and View in Library, while current-tab retargeting preserves the existing inspector state and refreshes the active packet identity fields.
- Packet tab decks now present Home as the workspace tab label, use middle truncation for packet titles, and surface desktop hover metadata so long packet titles and revision context stay discoverable without widening the deck.
- Home Export now includes an explicit Cancel reset path for packet-scoped export so preloaded packet targets can be cleared without leaving the Export workspace or disturbing the separate full local store export surface.
- Search preview caps and category browsing are now separated: All remains an 8-result grouped preview surface, while focused Direct, Name, and Text views page through larger result sets with server-backed 25-item pages.

2026-05 Dynamic scope graph consumer pass 1

- Home locality is now relation-first in the mutation corridor: canonical writes use `relation.residence.add` and produce `Relation(subtype: residence)` without automatically minting supporting claims.
- Legacy `residence.claim.set`, legacy `Claim(residence)`, and legacy `parent_scope` ancestry remain readable only through named compatibility adapters and projections rather than through mixed main-path shell logic.
- Mounted scope projection now prefers packet-native structural relations such as `defaultancestryparent` and `definedbylocation`, while also exposing richer packet-backed shell metadata for later UI graph and dashboard work.
- OWA home-locality projection now resolves from packet-native relations sourced through the forward `Action(subtype: initiative)` anchor path; relation requirements remain available for stricter policies but are not the default fresh-write wrapper.

2026-05 Scope graph hardening and packet-native scope writers

- Scope ancestry resolution now explicitly surfaces structural graph state, including `compatibilityparent`, `conflictingparents`, `cyclicancestry`, and `missingparent`, instead of silently treating all non-canonical cases as equivalent.
- Follow is now canonical and actor-only: new writes use `relation.follow.add` / `relation.follow.clear`, while shell cookie follow remain a compatibility read bridge only.

- Association is now relation-first in the mutation corridor: canonical writes use `relation.association.add` / `relation.association.clear`, keep the supporting self-claim layer, and mount associated scopes as related mounted scopes rather than leaving them as badge-only side state.
- The first packet-native shell UI pass keeps the current rail layout but now groups scope context into Home, Associated, Followed, and Discoverable sections, roots the home trunk at You, and routes follow/associate sidebar actions through the canonical mutation corridor with a refetched shell projection afterward.
- `locality.path.create` now emits locality Element packets, canonical defaultancestryparent relations, provisional Location(subtype: region) packets, and definedbylocation relations in one signed packet batch, while `parent_scope` remains a named compatibility mirror only.

2026-05 shared packet-card actions and focus

- Dashboard preview lists and focused packet panels now share reusable packet action menus instead of route-local action clusters.
- Dashboard preview surfaces can focus one packet at a time locally, letting the focused presentation reuse the same packet action vocabulary without inventing a separate packet-detail route.
- Compact tooltip-capable icon badges are now part of the shared Nexus packet-card language instead of being dashboard-only decoration.

2026-05 sidebar follow-up polish

- The Home scope trunk now renders broadest-to-smallest, with You at the personal end rather than as the visual root.
- Sidebar rows no longer depend on inline badge clutter or left-side tree graphics for their primary meaning.
- Compact side overflow menus are now the active scope-row action pattern, with section placement carrying most of the relationship signal.

2026-05 Explorer truth clarification

- Packet Explorer Search, Import, and Export are live and should be treated as current product truth.
- Explorer remains read-only for packet mutation.
- Verification is now live for local packet validation, import reporting, and Explorer verification surfaces, while richer peer trust, provenance weighting, and broader report semantics remain later seams.

2026-05 verification chapter 1

- Report is now a real packet type in code, with `verificationreport` and `importreport` as the first live subtypes.
- The local runtime now bootstraps a normal signed validator identity and uses it to sign packet verification and import reports instead of introducing a privileged core-signature bypass.
- Packet Explorer now has a dedicated Verification rail, while dashboard and shared packet action menus can Validate, Revalidate, and View verification.
- Import now supports Validate first, Validate after, and Don't validate, with validation reports and packet-level verification summaries recorded through runtime services and cached locally for truthful UI projection.

2026-05 verification hardening and bundle direction lock-in

- Packet-signature verification now accepts a narrow explicit set of identity-bearing signer packets, including `Element(kind: person)` and `Element(kind: service)`, so the local validator stays a real service identity instead of a fake person-only compatibility hack.
- Verification summaries are now revision-anchored: local cache rows record the exact target revision and digest they verified, and current verified state only applies while that preferred revision still matches.
- Unknown signer status is now intentionally truthful: a packet can remain signed while still being unverifiable on the local node when the signer packet is unavailable, instead of being mislabeled as cryptographically valid.
- Explorer Import now exposes recent `import_report` history and richer artifact metadata through the existing report packet type rather than introducing an interim import-history packet type.
- The future bundle direction is now locked in at the roadmap level: Bundle will become a first-class packet type later, rebundling will create a new bundle packet rather than revising someone else's bundle, bundle history should behave as a lineage graph, and legacy bundle JSON remains a runtime transport artifact rather than a core packet-compatibility type.

2026-05 verification tidy polish

- Explorer verification now surfaces freshness explicitly: the current preferred revision, the revision targeted by the latest local report, and whether that report is current, stale, or absent.
- Import History now speaks honestly about its current model: it shows the latest local import report per artifact identity rather than pretending to be a full append-only event ledger.

- The next major implementation chapter remains locality UX and geo standardization; a dedicated validation workflow screen and first-class Bundle packets stay intentionally unsupported for later work.

2026-05 locality UX pass 1

- `/nexus/locality/create` now has a true non-mutating Review path seam instead of pretending the write corridor itself is the review step.
- Top-level locality search now activates the existing `create_candidate` handoff so missing places can seed the broad-to-narrow builder directly, including from the Enter key path.
- Duplicate warnings in locality review are now actionable through Use existing, Edit name, and Create anyway rather than appearing only as passive blocker text.
- Use as home locality is now an explicit shared toggle for both selecting existing localities and creating new ones.
- The home-branch inclusion checklist is now visible during review as a UI-only preview affordance, while authoritative storage and projection of selected included scopes remains unsupported to a later locality pass.

2026-05 locality foundations phase 2A

- Locality planner semantics are now ordered-path first: the runtime interprets locality preview and create input as a broad-to-narrow path, and sparse paths no longer need to fill every legacy ladder slot.
- The current locality type picker is now real runtime input instead of decoration: path rows can carry `LocalityScopeDescriptor` data, with safe fallback descriptors derived from current legacy buckets when callers omit them during rollout.
- New locality writes now persist descriptor metadata inside linked `Location.spatialpayload.scopeddescriptor`, while legacy nation | region | city | district values remain compatibility buckets rather than the forward locality ontology.
- Locality search and duplicate projection now preserve Unicode scripts, normalize accents safely for matching, and keep useful legacy ASCII-style aliases where that compatibility bridge is feasible.
- Home-tree inclusion persistence and projection fields remain intentionally unsupported to the next locality foundations pass rather than being bundled into descriptor and ancestry work.

2026-05 locality hardening pass 2A.5

- Locality preview and create now reject decreasing legacy compatibility bucket order such as city -> nation, while still treating ordered path ancestry rather than the ladder itself as the real graph truth.

- Descriptor reads now validate `hierarchy_system` against the allowed runtime enum so malformed imported Location metadata falls back safely instead of being trusted as-is.
- Top-level locality search can now still route into creation when same-name results exist in a different branch, preventing one existing Ontario from suppressing a new Ontario under another parent path.
- The create page now clears stale carried-over target rows more carefully when the target type changes across legacy buckets, and successful create/reuse completion resets the create builder before the success modal appears.
- Trust home-locality presentation has been tightened into a smaller onboarding-friendly card without changing the underlying home-locality actions or relation-first write path.

2026-05 locality runtime catch-up chapter

- Locality confirmation now routes through one composite `locality.graph.apply` mutation intent above `locality.path.create`, so structural locality writes, selected home-locality changes, scope associations, follow, and display preferences are coordinated together instead of being chained from the client.
- Partial success is now explicit at that runtime seam: structural locality planning and packet writes remain phase one, while relation and display-preference writes report their own outcomes without pretending the whole flow is all-or-nothing.
- Actor-to-scope relationship reads are now centralized through one runtime controller that treats canonical residence, association, and follow relations as the main truth, while preserving guest and compatibility follow behavior as an explicit fallback bridge.
- `main` is a visible-scope preference rather than a relation, and associated/followed parent-context display toggles are persisted alongside it through the current claimed-actor preference bridge.
- The generic scope-graph projection now returns server-projected home, associated, followed, main, and discoverable sections for the sidebar, and OWA-specific initiative-anchor relation-policy lookup has been moved out of the generic graph core into a narrower adapter layer.

2026-05 Preference.element packet write corridor

- Preference is now enrolled in the canonical packet ontology as a replaceable packet type, with `Preference.element` carrying actor-owned scope-display preferences.
- Claimed actor scope-display writes now create live `Preference.element` packet revisions and also update the legacy runtime preference table as a compatibility cache.
- Scope-display reads prefer the latest active `Preference.element` packet and fall back to the legacy table when no packet exists; guest preferences remain cookie/session compatibility state.

- Claimed shell preference reads now resolve the actor from the authenticated session; actor query mismatches use guest projection instead of reading another actor's private preference packet.
- Preference.element now includes an interface.shell_chrome section for navigation mode, theme mode, and UI density so remaining shell-interface preferences can move into the same body without redefining the packet shape.
- runtime/nexus/server/packet-runtime-master-handler.ts is the first generic runtime connector corridor: routes can request a connector id, the handler resolves the packet definition and mutation action plan, then dispatches to local trusted connector code instead of hardcoding route-specific packet writes.
- preference.element.interface.set was initially enrolled in that master handler as a live bridge. It wrote Preference.element.value.interface.scopedisplay, could preserve or update interface.shellchrome, and kept the legacy scope-display table as a compatibility cache. The later fortress promotion pass moves claimed writes to preference.element.set and keeps this connector runtime-ready.
- The manifest definition fortress bridge remains non-executing and descriptor-only. The new runtime master handler still runs trusted local connector code; it does not execute arbitrary behavior from imported definitions.

2026-05 policy/dependency semantic authority

- Policy current schema now includes nullable defaultpolicy and governancepolicy sections so default inheritance and governance readiness are packet-backed before reseed design.
- Dependency refs in Definition parts and workflow plans now resolve through packet dependency semantics, Policy semantics, operation ontology entries, trusted workflow resolvers, or explicit local engine contracts instead of loose runtime-only strings.
- The seeded OWA Action(subtype: initiative) is now the policy/default anchor and links to default-inheritance and governance-baseline policies.

2026-05-20: Canonical subtype reset before reseed

- Active packet canon is pruned to Definition, Bundle, Element, Location, Role, Claim, Relation, Report, Proposal, Reaction, Decision, Action, Discussion, Policy, and Preference.
- Cause, split discussion types, split initiative/work types, Signal, Minutes, Artifact, and other unused alpha types are removed from fresh packet canon.
- Fresh active packet bodies use top-level body.subtype as the packet classifier. Old top-level classifier names are treated as alpha archive shapes, not fresh write compatibility obligations.

- Reseed continuity will preserve identity/key continuity by default; stale packet-id relations, home locality, follow, associations, main-tree scope IDs, and old discussion placement are not carried forward by default.

2026-05 final pre-reseed wrap-up

- Live fresh writes no longer accept `association.claim.set` or `residence.claim.set`; canonical association and home-locality writes now use relation-first mutation intents only.
- Legacy claim/home-locality material remains compatibility-readable and importable, but it is no longer a live mutation corridor endpoint.
- A final pre-reseed readiness report now inventories canonical write intents, compatibility-only surfaces, OWA default anchors, required seed policies, discussion defaults, canonical definition packet types, and out-of-scope packet types for the separate reseed design pass.

2026-05 definition packetization and Preference fortress promotion

- Active manifest definition parts now produce schema-validated canonical Definition packet envelopes, and those envelopes are grouped into one canonical `Bundle.packet_set` definition profile inventory for reseed readiness.
- The seeded definition profile audit compares packet material back to the core manifest and fails closed on missing parts, unexpected parts, stale profile metadata, duplicate bundle refs, or digest drift.
- Stored Definition and Bundle packet material is now reseed truth material, but execution remains trusted-local; imported definition packets may describe schemas, operations, policies, dependencies, planners, and builders, but they cannot introduce server code.
- Definition, Bundle, and Preference are now canonical packet types with body schemas, compatibility entries, builder support, pipeline inventory coverage, and packet-store validation like other active packet types.
- Claimed shell preference writes now run through the standard signed mutation prepare/finalize corridor as `preference.element.set`, preserving projected responses, actor proof/session checks, same-value no-op behavior, and legacy cache sync. `/api/nexus/shell-preferences` remains guest compatibility state only.
- The direct `preference.element.interface.set` runtime connector is retained as a definition/internal comparison bridge rather than the live claimed-write corridor.

2026-05 Nexus discussions UI decomposition

- Nexus discussions began moving route-local UI into `app/components/nexus/features/discussions/*` while preserving `src/app/nexus/discussions.tsx` as the route/controller owner for query state, loading, mutations, auth gates, and reply branch state.

- Extracted discussion feature components now cover feed post cards, root post cards, vote/reply-count pills, recursive reply trees, and post/reply composers.
- Discussion loading adoption remains visual-scope owned: feed load-more, root replies, reply branches, vote pills, and composers can mount scoped loading boundaries without packet-action registry or runtime coupling.
- Candidate universal pieces such as composer shells, vote/reaction skeletons, and tree rails remain feature-local until another Nexus surface proves the same reusable shape.

2026-05 Nexus discussions workspace panel extraction

- Discussion feed, thread, and post workspace sections now have feature-local panels under `app/components/nexus/features/discussions/*`, including a thread toolbar, while the route keeps router, auth, loading, mutation, and reply-state ownership.
- The panel props are explicit callbacks and display state rather than runtime services or router hooks, keeping the feature components ready for later generic promotion if another surface needs the same workspace/feed/thread/composer skeletons.
- Scoped loading boundaries remain attached to visual regions such as feed, root replies, reply branches, votes, and composers; operation semantics still stay outside UI.

2026-05 Nexus sidebar feature extraction

- Nexus sidebar rail/menu UI began moving into `app/components/nexus/features/sidebar/*` while `nexus-sidebar.tsx` remains the shell controller for router, auth, preference persistence, rail animation, and scope mutation state.
- Extracted sidebar feature components now cover rail toggles, guest avatar, preference switches, current context card, function menu content, scope section headers, grouped scope rows, scope action menus, and scope menu content.
- Scope follow, association, and main-tree visibility actions now use caller-owned loading scopes such as `sidebar:scope-follow:<scopeId>` and block the affected scope row/menu region without coupling loading to packet/runtime semantics.
- Candidate universal pieces such as rail toggles, compact nav rows, preference switches, grouped sections, scope/action rows, and anchored compact menus remain feature-local until another Nexus surface proves the same reusable shape.

2026-05 Nexus locality create UI decomposition

- Nexus locality create UI began moving into `app/components/nexus/features/locality/*` while `src/app/nexus/locality/create.tsx` remains the controller for params, auth gates, draft persistence, graph/path state, API calls, mutation handlers, and modal state.
- Extracted locality feature components now cover the search panel, builder panel, parent/type/create-kind dialogs, selected-result and remove confirmation dialogs, outcome dialogs, graph rows, and preview panel.

- Locality loading remains visual-scope owned: search results, parent picker results, graph row results, preview, create-path, and set-home actions use caller-owned loading scopes without coupling UI to packet/runtime semantics.
- The obsolete level-row ladder builder path was removed after extraction; the active create flow is the graph-based builder, with route/controller ownership preserved.